VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
UNIVERSITY OF INFORMATION TECHNOLOGY
FACULTY OF INFORMATION SYSTEMS

TECHNICAL REPORT

Superstore Business Intelligence Final Project

An End-to-End BI Solution: ETL, Data Warehouse, OLAP Cube, MDX Analytics, Power BI Dashboard, and Python Data Mining

Course: INTELLIGENT DECISION SUPPORT SYSTEMS

Lab Instructor: Nguyen Thi Kim Phung

Student: Vo Thanh Danh

Student ID: 22520201

Class: **MSIS4263.Q21.CTTT**

HO CHI MINH CITY, JUNE 2026

Executive Summary

This report documents an end-to-end Business Intelligence solution built on the Superstore sales dataset (9,994 order-line records, 21 columns) using the Microsoft BI stack supplemented by Python data mining. The solution covers five integrated layers: ETL pipeline design and execution, dimensional data warehouse construction, OLAP cube deployment and MDX analysis, interactive reporting in Power BI and Excel, and predictive/segmentation data mining.

The data warehouse (`SuperstoreDW`) adopts a star schema with one central fact table (`FactSales`) and five dimension tables (`DimDate`, `DimProduct`, `DimCustomer`, `DimGeography`, `DimShipMode`). The project does not use a separate staging database; CSV rows are first loaded into `SuperstoreDW.dbo.stg_Orders`, an internal landing table inside the main warehouse database. The ETL process, implemented in SQL Server Integration Services (SSIS), then transforms the internal landing rows into the warehouse structure; reconciliation checks confirm that row counts, total Sales, and total Profit match between the landing table and the fact table, with zero orphan foreign keys.

The SSAS Multidimensional cube (*Superstore DW*) exposes the star schema as a browsable OLAP cube enriched with two calculated members (*Profit Margin* and *Average Delivery Days*). Ten analytical business questions are answered in parallel through MDX queries, Power BI dashboard pages, and Excel PivotTables connected to the cube via OLAP.

The data mining component applies two complementary algorithms. A supervised Decision Tree classifier predicts whether each order line is profitable or loss-making, achieving 94.40% test accuracy and a macro F1-score of 90.33%, well above the 80.63% majority-class baseline. The strongest extracted rule— `Discount > 25% and Category != Technology` predicts a loss with 99.37% precision on 318 test records. An unsupervised K-Means model segments 793 customers by RFM behavior into four clusters (Silhouette score 0.3554): VIP / Champions (64 customers, highest monetary value $9,479.5), Trung thanh / Active (298 customers), Pho thong / Trung binh (335 customers), and Roi bo / Lost (96 customers with 559.5 days mean recency).

The central business finding is that deep discounting outside the Technology category is the primary driver of loss-making transactions, while a small cohort of VIP customers generates disproportionate revenue and requires dedicated retention investment.

Keywords: business intelligence, data warehouse, star schema, SSIS, SSAS, MDX, Power BI, decision tree, K-Means, RFM, Superstore.

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1 Introduction

1.1 Project Objective

The objective of this project is to design and implement a complete, production- grade Business Intelligence pipeline for a retail analytics context. The pipeline transforms raw transactional data from the Superstore dataset into actionable business insights through a sequence of interoperating components: data extraction and loading, dimensional modelling, OLAP cube construction, multi-tool analytical reporting, and machine-learning-based data mining.

BI systems of this architecture follow the well-established three-tier paradigm described by Kimball and Ross [1]: a source tier (flat-file CSV), an integration and warehouse tier (SSIS ETL and SQL Server star schema), and a presentation tier (SSAS cube, Power BI, Excel, and Python). The solution in this report adheres to this paradigm and complements it with a data mining layer that produces predictive and segmentation outputs not available from OLAP alone.

1.2 Business Questions

The system is designed to answer ten core analytical business questions:

1. How do Sales and Profit evolve by Year and Quarter?
2. Which US States generate the highest Sales and Profit?
3. Which ten products rank highest by Sales?
4. How do Sales and Profit compare across Product Sub-Categories?
5. What is the Profit Margin by Customer Segment?
6. How does Discount level affect Profit across Categories?
7. How does Sales volume and order count vary by Ship Mode?
8. What is the Year-over-Year Sales growth rate?
9. Which ten customers generate the most Sales?
10. What is the average delivery time by Region and Ship Mode?

1.3 Dataset Description

The source is `Sample - Superstore.csv` [2], a widely used retail analytics benchmark dataset. It contains **9,994 rows** at order-line grain (one row per product within one order) and 21 columns. The primary analytical measures are `Sales`, `Quantity`, `Discount`, and `Profit`. Descriptive attributes cover order and shipping dates, ship mode, customer identity and segment, geography (Country, Region, State, City, Postal Code), and product hierarchy (Category, Sub-Category, Product Name). The dataset uses Windows-1252 encoding, which requires explicit handling during SSIS Flat File source configuration and in the Python data mining script.

1.4 Solution Architecture

The end-to-end data flow is summarised below and illustrated in Figure 1:

`CSV` → `SSIS ETL` → `SuperstoreDW.dbo.stg_Orders` → `Star Schema (FactSales + 5 Dims)`
→ `SSAS Cube` → `MDX / Power BI / Excel` → `Python Data Mining`

The submitted pipeline does not use a separate staging database. The CSV is loaded into `SuperstoreDW.dbo.stg_Orders`, an internal landing table inside `SuperstoreDW`, before transformation into dimensions and `FactSales`. SSAS and Power BI consume only the curated dimensional model composed of `FactSales` and the dimension tables.

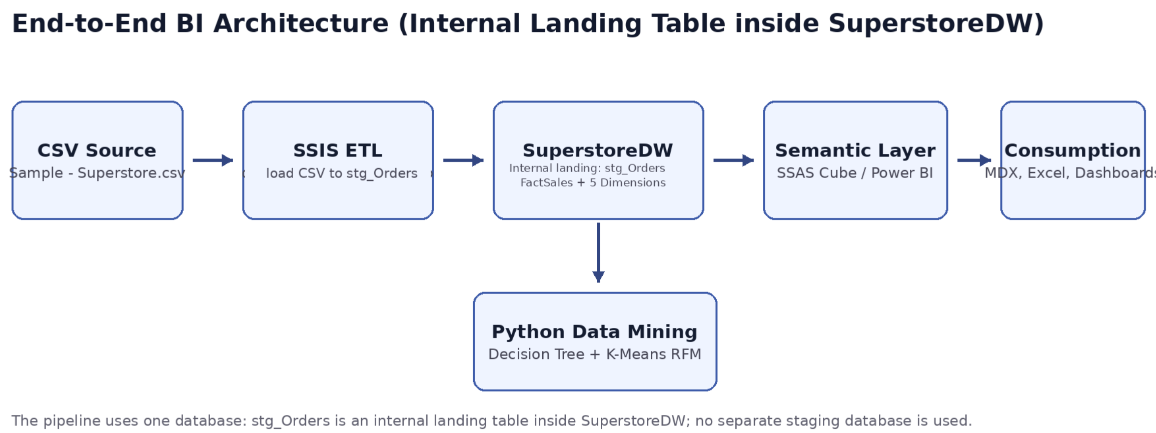


Figure 1: End-to-end architecture of the Superstore BI solution. The pipeline uses `SuperstoreDW.dbo.stg_Orders` as an internal landing table inside `SuperstoreDW`; it does not create a separate staging database. SSAS, Power BI, and Excel consume the analytical schema.

1.5 Tools and Technologies

Table 1: Technologies used in the Superstore BI solution.

Component	Tool / Technology
Database engine	SQL Server 2022 Developer Edition
ETL	SQL Server Integration Services (SSIS), Visual Studio 2022
OLAP	SQL Server Analysis Services (SSAS) Multidimensional mode
Query language	SQL and MDX
Dashboard	Power BI Desktop
Pivot analysis	Microsoft Excel (OLAP connection to SSAS cube)
Data mining	Python 3: `pandas`, `scikit-learn`, `matplotlib`
DB management	SQL Server Management Studio (SSMS)

2 SSIS ETL Process

2.1 Source Analysis

The source dataset contains 21 columns, summarised in Table 2. The file uses Windows-1252 encoding, which must be set explicitly in the SSIS Flat File Connection Manager to prevent character substitution in product names and city names. The column `Postal Code` has sparse missing values for a small number of records; these are carried through to the `DimGeography` table as `NULL` without requiring row removal, as the geographic hierarchy remains resolvable through State and City.

Table 2: Source dataset columns and their warehouse mapping.

Column	Description	Warehouse destination
Row ID	Sequential row identifier	Dropped (not loaded)
Order ID	Degenerate order identifier	`FactSales.OrderID`
Order Date	Date of order placement	`DimDate` (OrderDateKey)
Ship Date	Date of shipment	`DimDate` (ShipDateKey, role-playing)
Ship Mode	Shipping method	`DimShipMode`
Customer ID	Unique customer identifier	`DimCustomer`
Customer Name	Full customer name	`DimCustomer`
Segment	Customer segment	`DimCustomer`
Country	Order country	`DimGeography`
City	Order city	`DimGeography`
State	Order state	`DimGeography`
Postal Code	Postal code (sparse nulls)	`DimGeography`
Region	Sales region	`DimGeography`
Product ID	Unique product identifier	`DimProduct`
Category	Product category	`DimProduct`
Sub-Category	Product sub-category	`DimProduct`
Product Name	Full product name	`DimProduct`
Sales	Line revenue ($)	`FactSales.Sales` (measure)
Quantity	Units ordered	`FactSales.Quantity` (measure)
Discount	Fractional discount applied	`FactSales.Discount` (measure)
Profit	Line profit ($)	`FactSales.Profit` (measure)

2.2 Grain Definition

The warehouse preserves the **order-line grain**: each row in `FactSales` corresponds to one product line within one order, identical to the source CSV grain. This grain supports analytical aggregation by any combination of date, product, customer, geography, and shipping mode without loss of detail.

2.3 Single-Database Design

SSIS connects to a single database—`SuperstoreDW`—through the `OLEDB_DW` connection manager. The submitted design does not use a separate staging database. The package uses only

SuperstoreDW; the table SuperstoreDW.dbo.stg_Orders is an internal CSV landing table in the same database. Source records are then transformed into the warehouse structure and loaded into the dimensional tables and FactSales inside SuperstoreDW. This design simplifies deployment because the analytical model, cube source, Power BI model, and verification scripts all target one warehouse database.

2.4 ETL Package Structure

The main SSIS package (ETL_Main.dtsx) implements the following Control Flow sequence (Figure 2):

1. **Load source CSV into internal landing table:** The Flat File Source reads Sample - Superstore.csv with Windows-1252 encoding and loads the rows into SuperstoreDW.dbo.stg_Orders.
2. **Load DimDate:** The calendar dimension is pre-populated by the script sql/03_populate_dimdate.sql; SSIS does not re-generate it at runtime.
3. **Load dimension tables:** Distinct attributes from the internal landing table are loaded into DimProduct, DimCustomer, DimGeography, and DimShipMode.
4. **Load FactSales:** The fact-loading Data Flow reads from SuperstoreDW.dbo.stg_Orders, resolves surrogate keys from each dimension via Lookup transformations, derives the DeliveryDays column (ShipDate − OrderDate), and inserts the analytical measures into FactSales (Figure 4).

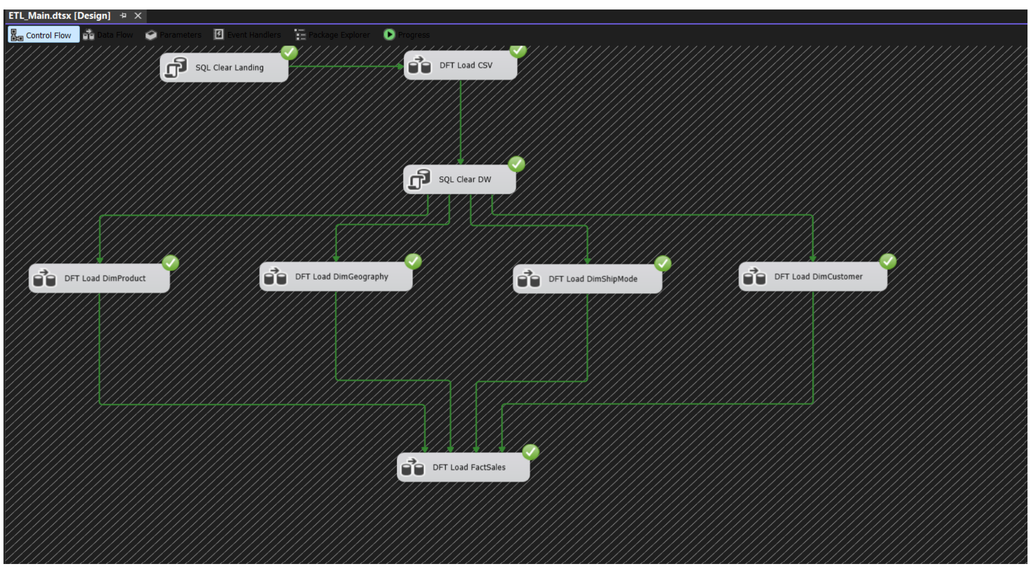


Figure 2: Control Flow of the SSIS package ETL_Main.dtsx. The package uses one DW connection and no separate staging database; all tasks complete successfully.

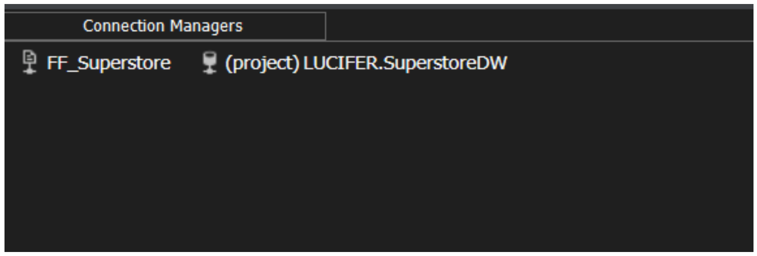


Figure 3: SSIS Connection Managers. Only `OLEDB_DW` connects to `SuperstoreDW`; there is no `SuperstoreStaging` connection manager. The Flat File connection points to the CSV source with Windows-1252 encoding.

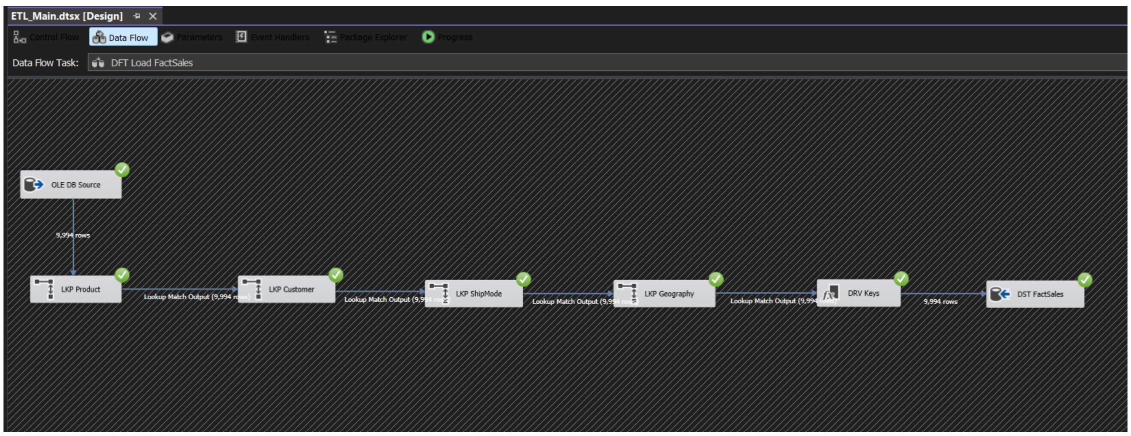
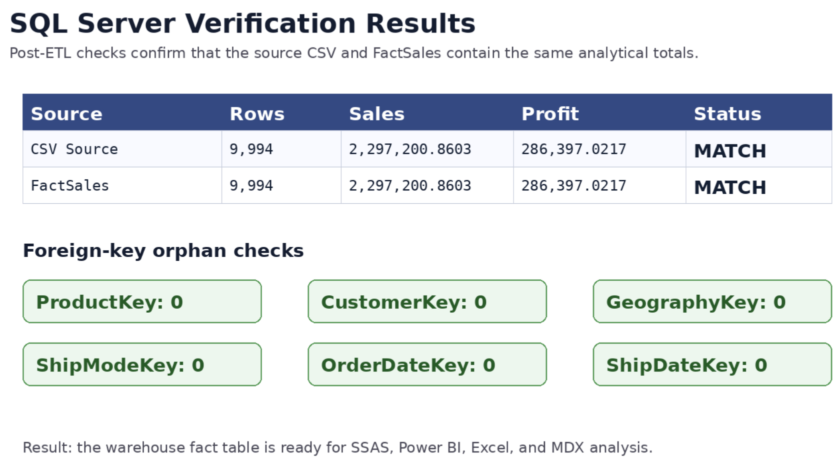


Figure 4: Data Flow task that populates `FactSales`. The source side is the internal landing table `SuperstoreDW.dbo.stg_Orders`; Lookup transformations resolve surrogate keys from each dimension, a Derived Column adds `DeliveryDays`, and the result is written to `FactSales`.

2.5 ETL Validation

After the package executes, `sql/06_verify.sql` performs three reconciliation checks (Figure 5):

- **Row count:** `SuperstoreDW.dbo.stg_Orders` and `FactSales` both contain exactly 9,994 order-line records.
- **Aggregate totals:** Total Sales and total Profit in `FactSales` match the corresponding landing-table aggregates to four decimal places.
- **Referential integrity:** All orphan foreign-key queries return zero rows, confirming that every fact record has a valid surrogate key in each dimension.



SQL Server Verification Results

Post-ETL checks confirm that the source CSV and FactSales contain the same analytical totals.

Source	Rows	Sales	Profit	Status
CSV Source	9,994	2,297,200.8603	286,397.0217	MATCH
FactSales	9,994	2,297,200.8603	286,397.0217	MATCH

Figure 5: Results of `sql/06_verify.sql`. Row counts, Sales totals, and Profit totals match between `SuperstoreDW.dbo.stg_Orders` and `FactSales`; the source side of the reconciliation is not a separate staging database, and all orphan-key counts equal zero.

3 Data Warehouse and SSAS Cube

3.1 Star Schema Design

The data warehouse uses a **star schema** (Figure 6) with one central fact table and five surrounding dimension tables. The design follows Kimball's guidelines for dimensional modelling [1]: facts capture measurable business events at the defined grain, dimensions provide the rich descriptive context needed for slicing, filtering, and drilling.

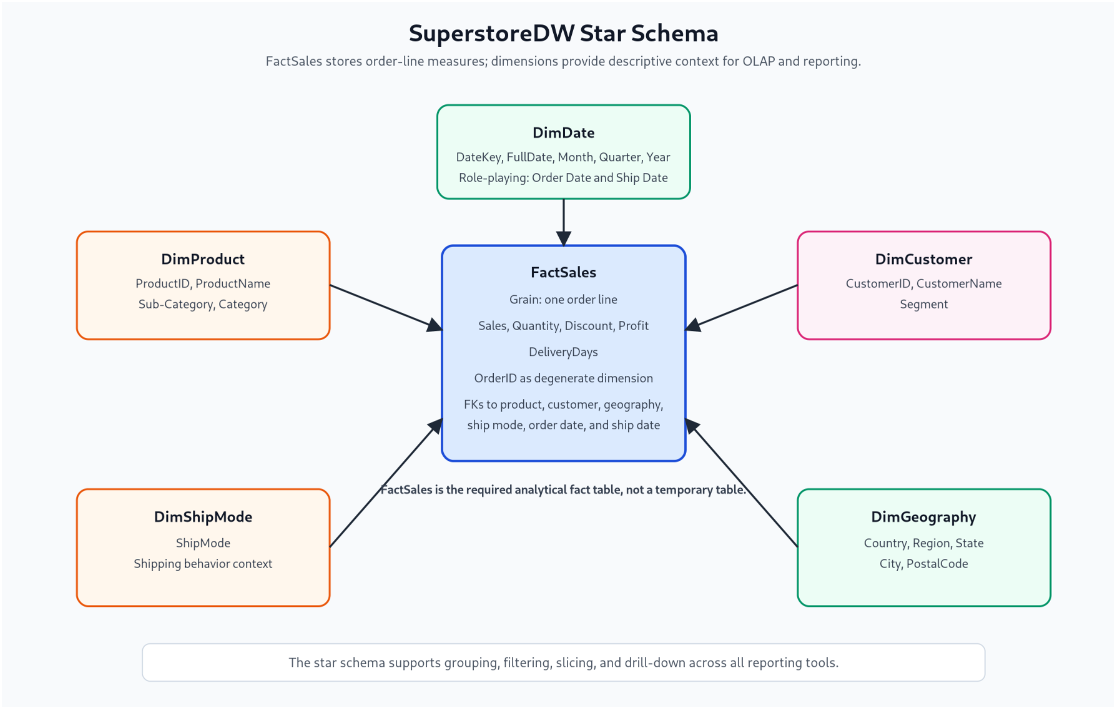


Figure 6: Star schema of the `SuperstoreDW` analytical layer. `FactSales` is the central fact table; `DimDate` serves as a role-playing dimension for both `OrderDateKey` and `ShipDateKey`.

3.1.1 Fact Table: FactSales

`FactSales` stores one row per order-line and carries six foreign keys (two referencing `DimDate` in different roles), a degenerate dimension (`OrderID`), and five measures:

Table 3: `FactSales` column definitions.

Column	Type	Description
FactSalesID	INT IDENTITY	Surrogate primary key
OrderDateKey	INT (FK)	Date of order placement; references `DimDate`
ShipDateKey	INT (FK)	Date of shipment; references `DimDate` (role-playing)
ProductKey	INT (FK)	References `DimProduct`
CustomerKey	INT (FK)	References `DimCustomer`
GeographyKey	INT (FK)	References `DimGeography`
ShipModeKey	INT (FK)	References `DimShipMode`
OrderID	NVARCHAR(20)	Degenerate dimension (no separate dim table needed)
Sales	DECIMAL(18,4)	Line revenue in USD
Quantity	INT	Units ordered
Discount	DECIMAL(9,4)	Fractional discount (0.0–1.0)
Profit	DECIMAL(18,4)	Line profit in USD (can be negative)
DeliveryDays	INT	ShipDate − OrderDate (derived in SSIS)

3.1.2 Dimension Tables

Table 4 summarises the five dimension tables and their key attributes. `DimDate` is the only dimension without an `IDENTITY` primary key: its `DateKey` uses the `yyyymmdd` integer format, which supports direct date arithmetic and makes date filtering intuitive.

Table 4: Dimension table summary.

Dimension	Key attributes	Rows
DimDate	DateKey (yyyymmdd), FullDate, Day, Month, MonthName, Quarter, Year, WeekdayName	Calendar-generated
DimProduct	ProductKey, ProductID, ProductName, SubCategory, Category	1,862
DimCustomer	CustomerKey, CustomerID, CustomerName, Segment	793
DimGeography	GeographyKey, Country, Region, State, City, PostalCode	631
DimShipMode	ShipModeKey, ShipMode	4

DimDate is populated by `sql/03_populate_dimdate.sql`, not by SSIS. Dimension row counts reflect the number of distinct attribute combinations in the source dataset.

Figure 7 shows the `SuperstoreDW` table list in SSMS, confirming the presence of all six tables.

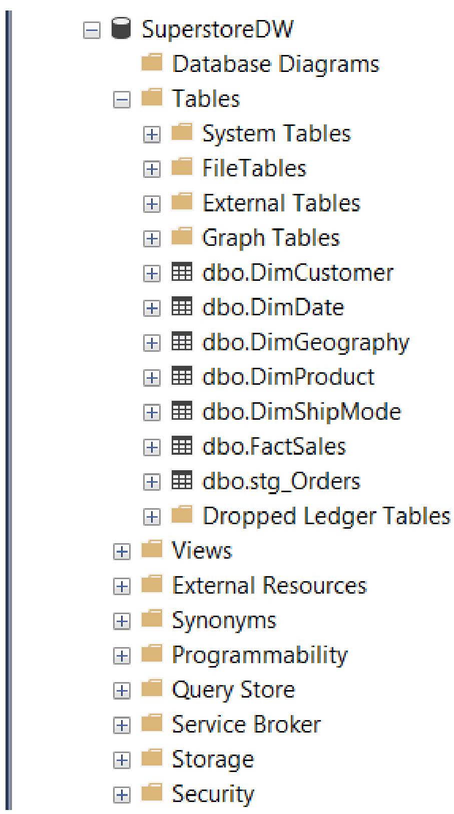


Figure 7: `SuperstoreDW` table list in SQL Server Management Studio, confirming that all five dimension tables and `FactSales` are present.

3.2 Star Schema vs. Snowflake Schema

A snowflake schema would normalise the product and geography hierarchies into additional sub-tables (e.g. separating Category and Sub-Category from `DimProduct`, or separating Country and Region from `DimGeography`). Figure 8 compares the two designs.

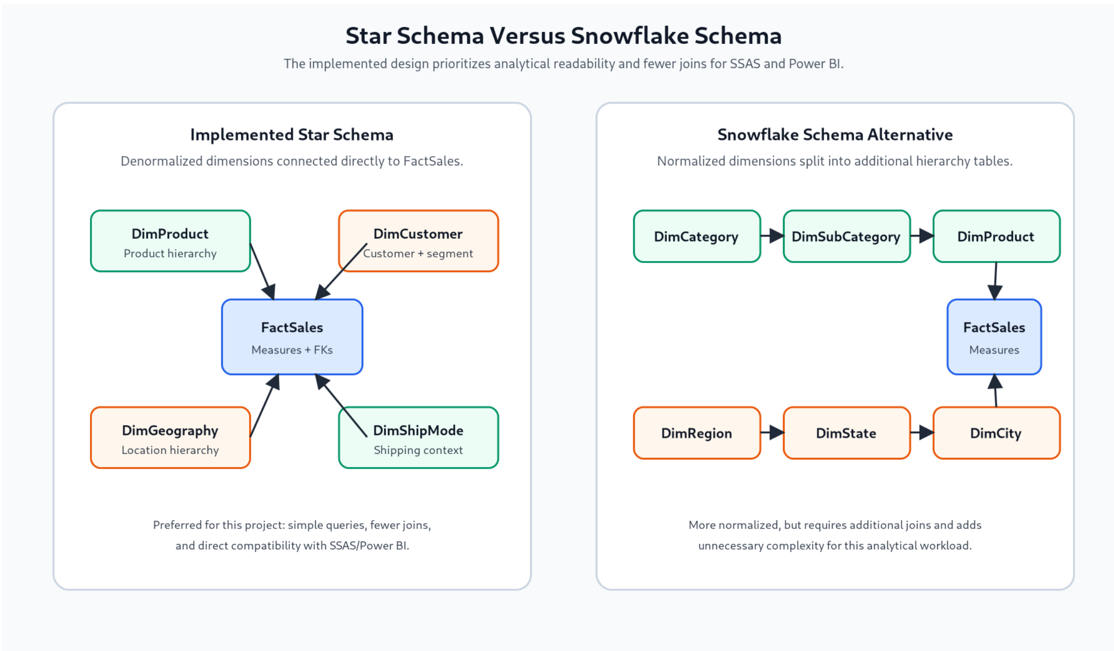


Figure 8: Star schema (left) vs. snowflake schema (right) for the Superstore DW. The star schema collapses the product and geography hierarchies into single wide dimension tables.

The project adopts the star schema for three reasons. First, fewer joins are required in OLAP and Power BI queries, reducing query complexity and improving dashboard responsiveness [1]. Second, SSAS Multidimensional builds hierarchies from attributes within a single dimension table more efficiently than from join paths across normalised tables. Third, the Superstore dataset is sufficiently small that the storage overhead of denormalisation is negligible. A snowflake schema would be preferable if storage cost or data update granularity were primary concerns—for example, when Category descriptions change independently of Sub-Category records.

3.3 SSAS Multidimensional Cube

3.3.1 Cube Configuration

The SSAS project (`SuperstoreSSAS`) deploys a Multidimensional cube named *Superstore DW* to the local Analysis Services instance configured in *Multidimensional and Data Mining Mode*. The data source (Figure 9) points to `SuperstoreDW`. The Data Source View (Figure 10) includes only the analytical star schema: `FactSales` and the five dimension tables.

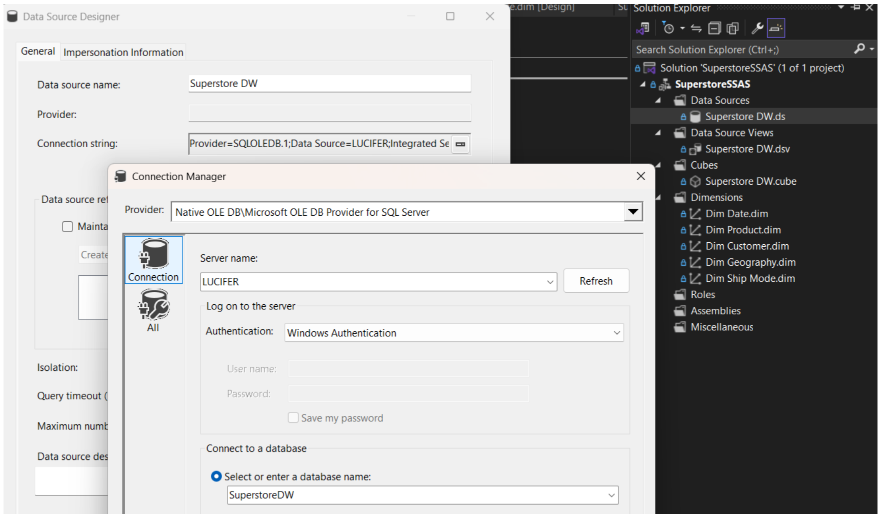


Figure 9: SSAS Data Source pointing to `SuperstoreDW`. The connection uses Windows Authentication and targets the local SQL Server instance.

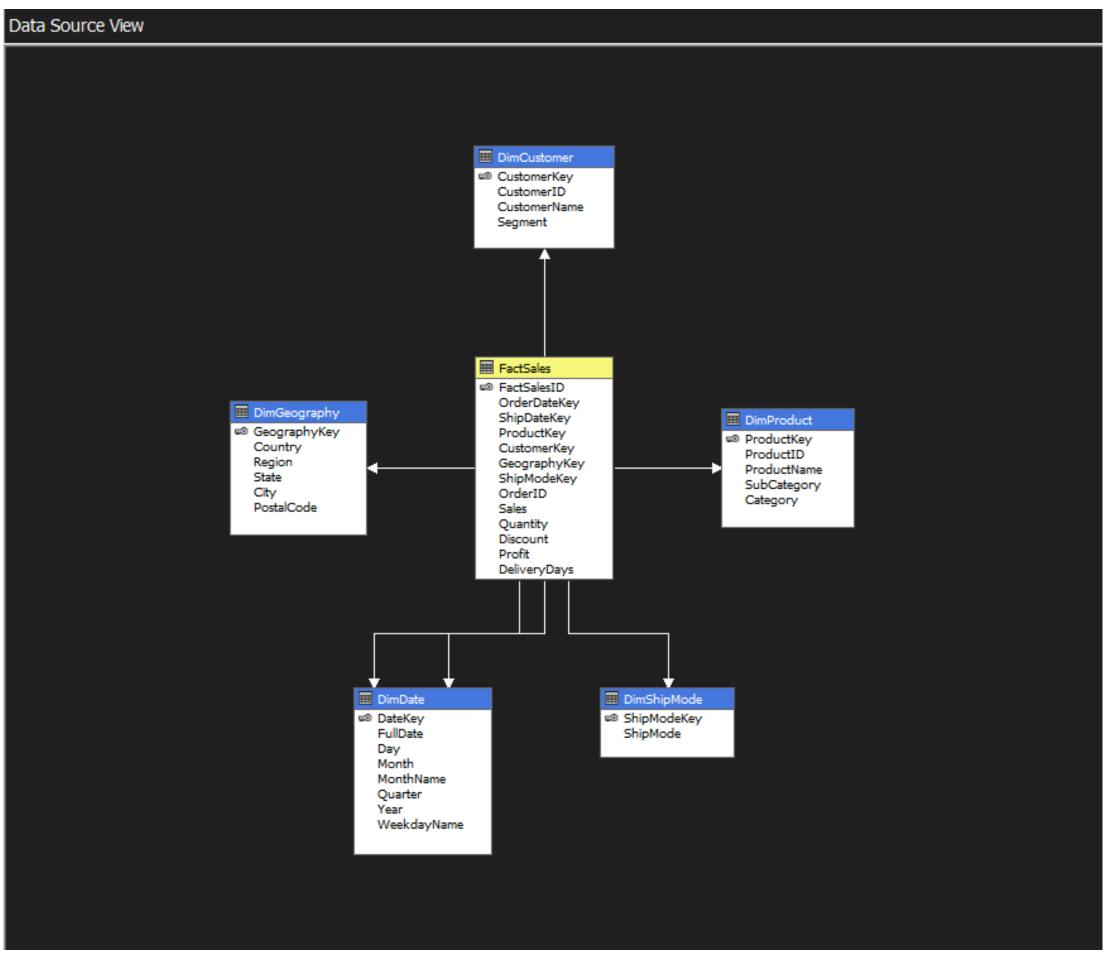


Figure 10: Data Source View in SSAS, showing `FactSales` at the centre with relationships to the five dimension tables. The role-playing date dimension appears as two named query aliases (`OrderDate` and `ShipDate`).

3.3.2 Dimensions and Hierarchies

The cube defines five user hierarchies aligned with the analytical questions:

- **Order Date / Calendar:** Year → Quarter → Month → Day
- **Dim Product:** Category → Sub-Category → Product Name
- **Dim Customer:** Segment → Customer Name
- **Dim Geography:** Region → State → City → Postal Code
- **Dim Ship Mode:** Ship Mode (flat, single level)

3.3.3 Measures and Calculated Members

The measure group contains six base measures sourced from `FactSales`: *Sales*, *Quantity*, *Discount*, *Profit*, *Delivery Days*, and *Fact Sales Count* (row count). Two calculated members are defined in the cube Calculations tab:

Listing 1: Calculated members defined in the SSAS cube.

```
-- Profit Margin
[Profit Margin] = IIF([Measures].[Sales] = 0, NULL,
                      [Measures].[Profit] / [Measures].[Sales])
-- Format: Percent

-- Average Delivery Days
[Avg Delivery Days] = IIF([Measures].[Fact Sales Count] = 0, NULL,
                          [Measures].[Delivery Days] /
                          [Measures].[Fact Sales Count])
```

Figure 11 shows the cube browser confirming that Sales and Profit are browsable by product category after deployment and processing.

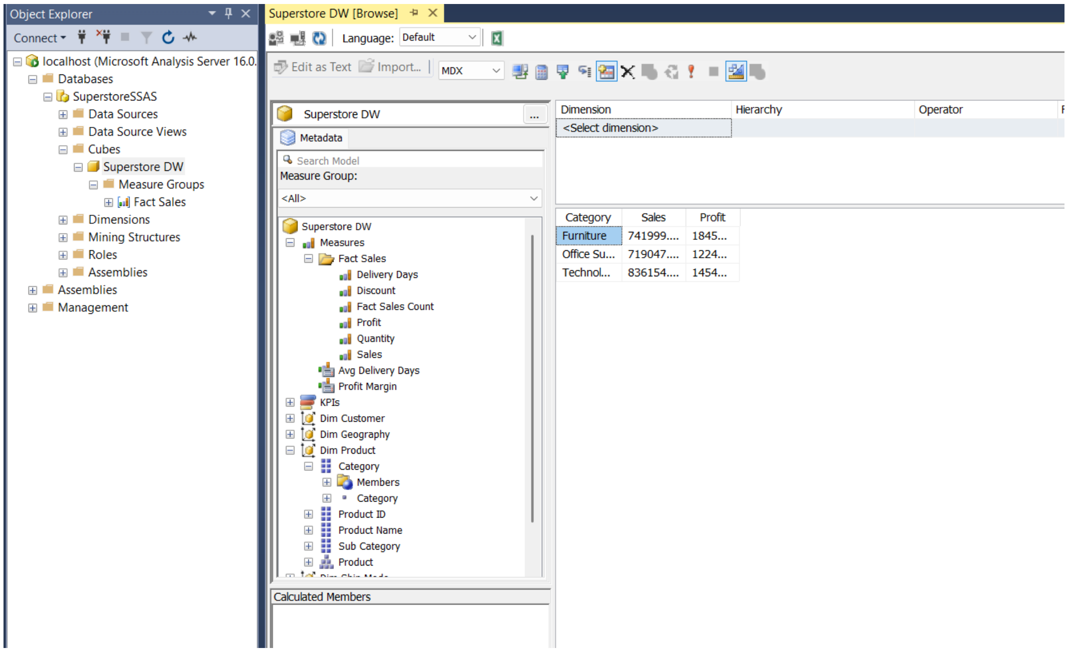


Figure 11: SSAS Cube Browser showing Sales and Profit by product category. The cube is fully processed and returning data.

4 Analysis and Reporting

4.1 MDX Analysis

Six representative MDX queries from `mdx/queries.mdx` are presented below. Each query was executed in SSMS against the deployed *Superstore DW* cube, and the result is illustrated with a screenshot.

4.1.1 Q1 — Sales and Profit by Year and Quarter

Listing 2: Q1: Sales and Profit by Year and Quarter.

```
SELECT { [Measures].[Sales], [Measures].[Profit] } ON COLUMNS,
NON EMPTY
HIERARCHIZE(
  DRILLDOWNLEVEL([Order Date].[Calendar].[Year].MEMBERS)
) ON ROWS
FROM [Superstore DW];
```

Interpretation: Sales grew from 2014 to 2017. The Q4 peak is consistently the strongest quarter in each year, driven by holiday-season demand for office supplies and technology products.

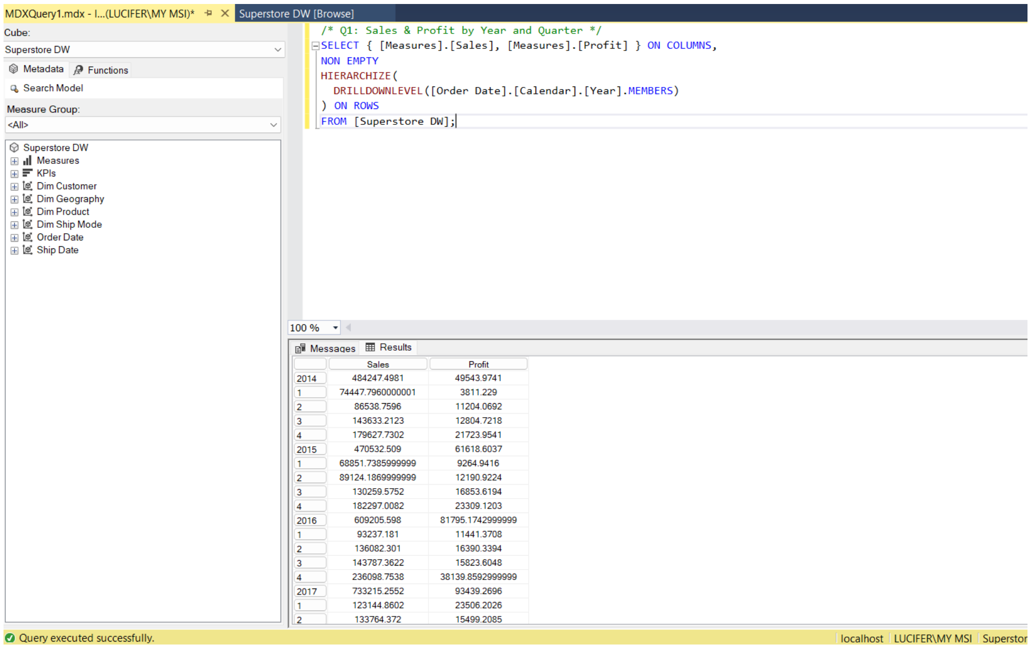


Figure 12: MDX Q1 result: Sales and Profit disaggregated by Year and Quarter. Q4 is the strongest quarter in each year.

4.1.2 Q3 — Top 10 Products by Sales

Listing 3: Q3: Top 10 Products by Sales.

```
SELECT [Measures].[Sales] ON COLUMNS,
TOPCOUNT([Dim Product].[Product].[Product Name].MEMBERS,
          10, [Measures].[Sales]) ON ROWS
FROM [Superstore DW];
```

Interpretation: High-ticket technology items (copiers, phones, and machines) dominate the top-10 list, indicating that Technology is the primary revenue driver despite the mixed profit outcomes shown in Q6.

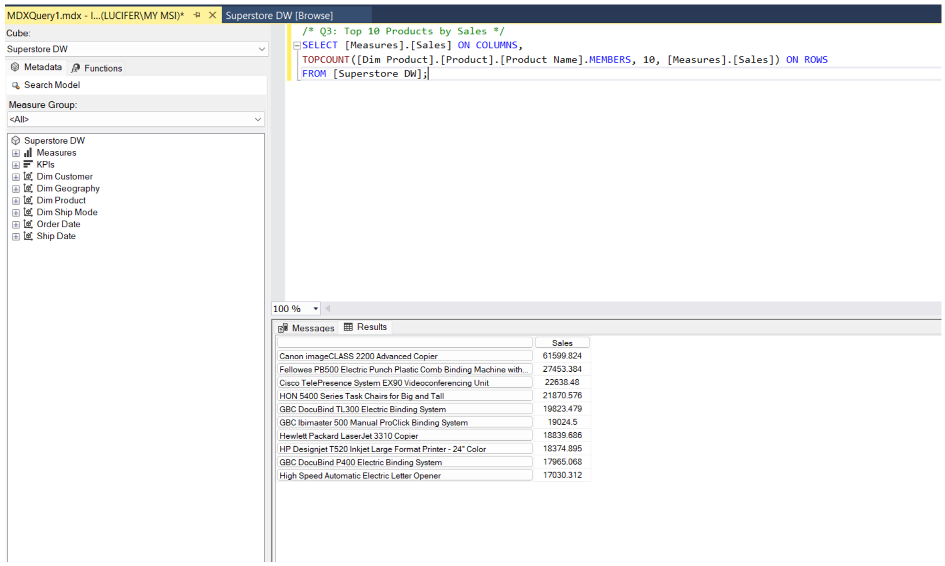


Figure 13: MDX Q3: Technology items dominate the top-10 revenue-generating products.

4.1.3 Q5 — Profit and Profit Margin by Customer Segment

Listing 4: Q5: Profit Margin by Customer Segment.

```
SELECT { [Measures].[Profit], [Measures].[Profit Margin] } ON COLUMNS,
[Dim Customer].[Customer].[Segment].MEMBERS ON ROWS
FROM [Superstore DW];
```

Interpretation: The Corporate segment produces the highest absolute Profit while the Home Office segment shows the strongest Profit Margin percentage. The Consumer segment has the highest Sales volume but the lowest margin rate, indicating room for discount discipline.

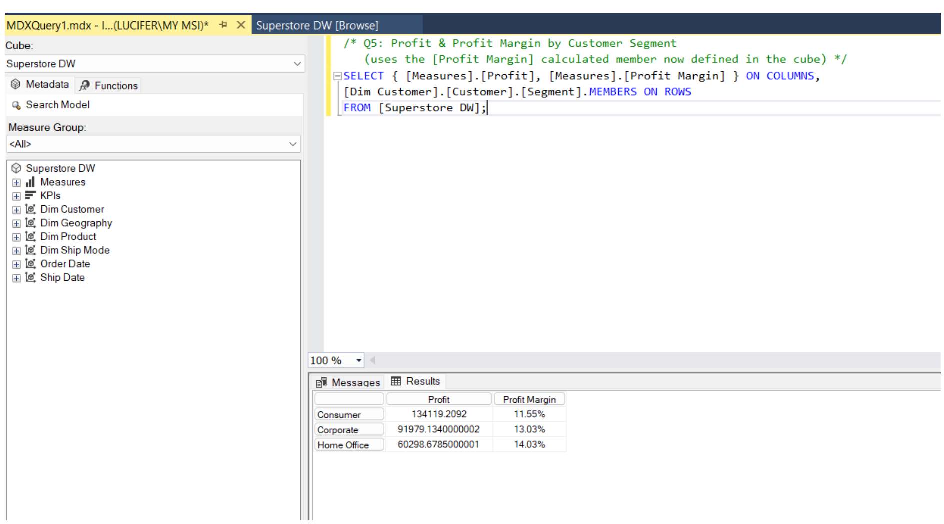


Figure 14: MDX Q5 result. Corporate leads on absolute Profit; Home Office shows the strongest Profit Margin.

4.1.4 Q6 — Discount vs Profit by Category

Listing 5: Q6: Discount and Profit by Product Category.

```
SELECT { [Measures].[Discount], [Measures].[Profit] } ON COLUMNS,
[Dim Product].[Product].[Category].MEMBERS ON ROWS
FROM [Superstore DW];
```

Interpretation: Furniture has a negative total Profit despite significant Sales volume, directly linked to its high aggregate Discount. This result is consistent with the Decision Tree rule extracted in Section 5: excessive discounting outside Technology destroys margin.

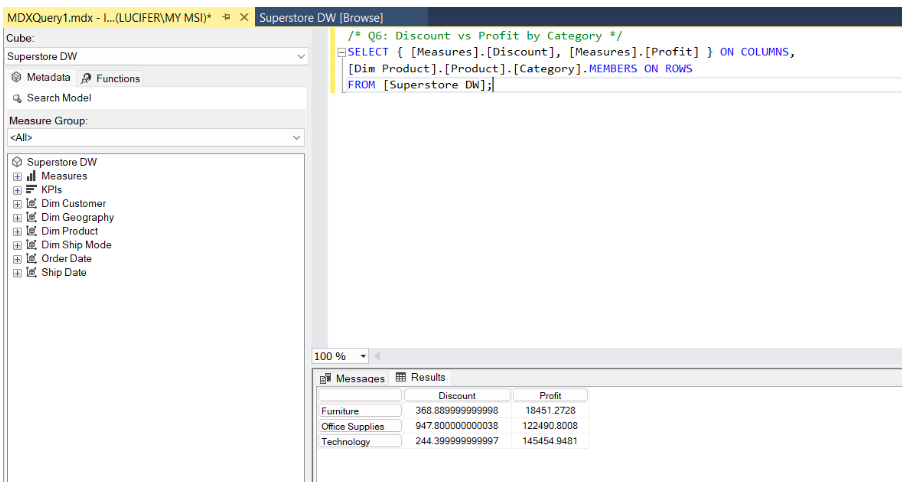


Figure 15: MDX Q6: Furniture shows a high aggregate Discount and negative Profit, confirming the discount–loss relationship.

4.1.5 Q8 — Year-over-Year Sales Growth

Listing 6: Q8: Year-over-Year Sales Growth with PARALLELPERIOD.

```
WITH MEMBER [Measures].[Sales PY] AS
  ([Measures].[Sales],
   PARALLELPERIOD([Order Date].[Calendar].[Year], 1,
                  [Order Date].[Calendar].CurrentMember))
MEMBER [Measures].[YoY %] AS
  IIF([Measures].[Sales PY] = 0, NULL,
      ([Measures].[Sales] - [Measures].[Sales PY])
      / [Measures].[Sales PY]),
  FORMAT_STRING = 'Percent'
SELECT { [Measures].[Sales], [Measures].[Sales PY],
         [Measures].[YoY %] } ON COLUMNS,
NONEMPTY([Order Date].[Calendar].[Year].MEMBERS,
         [Measures].[Sales]) ON ROWS
FROM [Superstore DW];
```

Interpretation: Sales grew positively in each year-over-year comparison across the dataset's time range, with the strongest growth occurring in the final year. The `PARALLELPERIOD` function correctly handles the prior-year reference for the first year (no previous year returns `NULL` for *Sales PY* and *YoY%*).

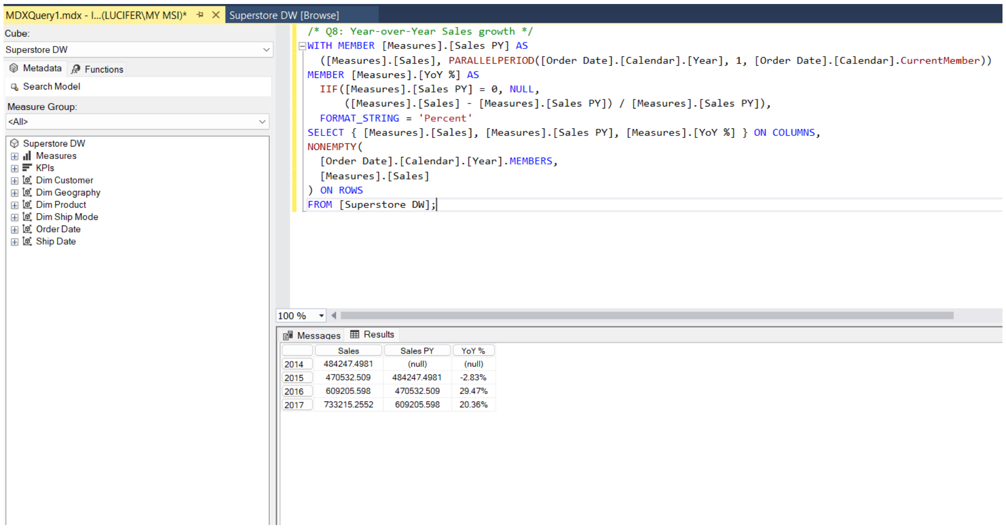


Figure 16: MDX Q8: Year-over-Year Sales Growth computed with `PARALLELPERIOD`. Sales increased in every period.

4.1.6 Q10 — Average Delivery Days by Region and Ship Mode

Listing 7: Q10: Average Delivery Days by Region and Ship Mode.

```
SELECT [Dim Ship Mode].[Ship Mode].[Ship Mode].MEMBERS ON COLUMNS,
[Dim Geography].[Geography].[Region].MEMBERS ON ROWS
FROM [Superstore DW]
WHERE ([Measures].[Avg Delivery Days]);
```

Interpretation: *Same Day* shipping achieves near-zero delivery days as expected. *Standard Class* averages approximately 5 days across all regions with low regional variance, suggesting that the shipping carrier's service level is consistent. *First Class* and *Second Class* occupy intermediate positions.

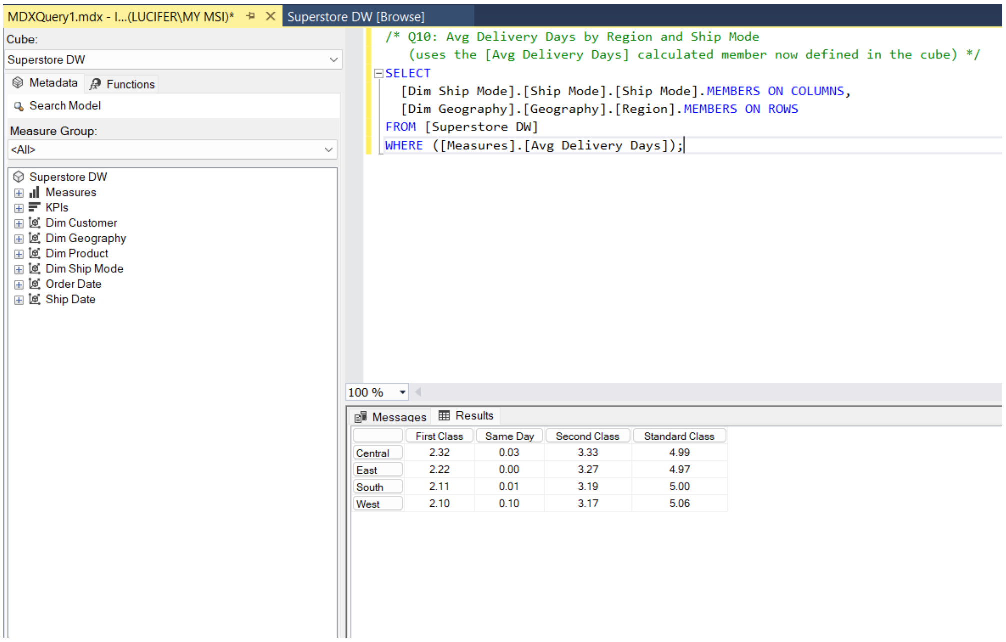


Figure 17: MDX Q10 result using the *Avg Delivery Days* calculated member. Standard Class averages approximately 5 days across all regions.

4.2 Power BI Reporting

Power BI Desktop connects to `SuperstoreDW` via an Import connection (Figure 18). Only the analytical tables (`Dim*` and `FactSales`) are imported. The model view (Figure 19) confirms the star-schema relationship layout with `FactSales` at the centre.

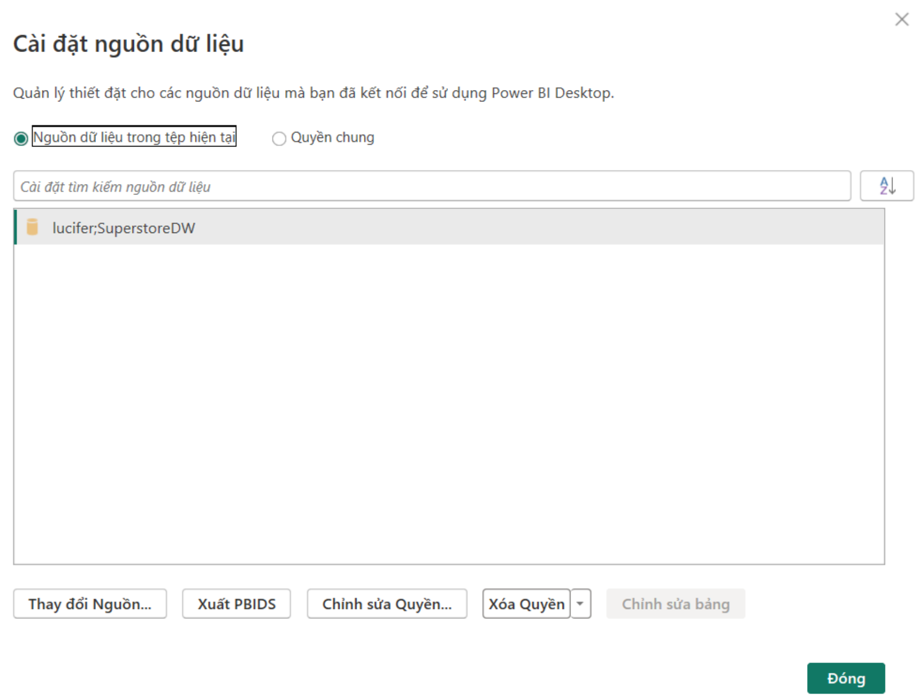


Figure 18: Power BI data source selection screen, connecting to `SuperstoreDW` in Import mode.

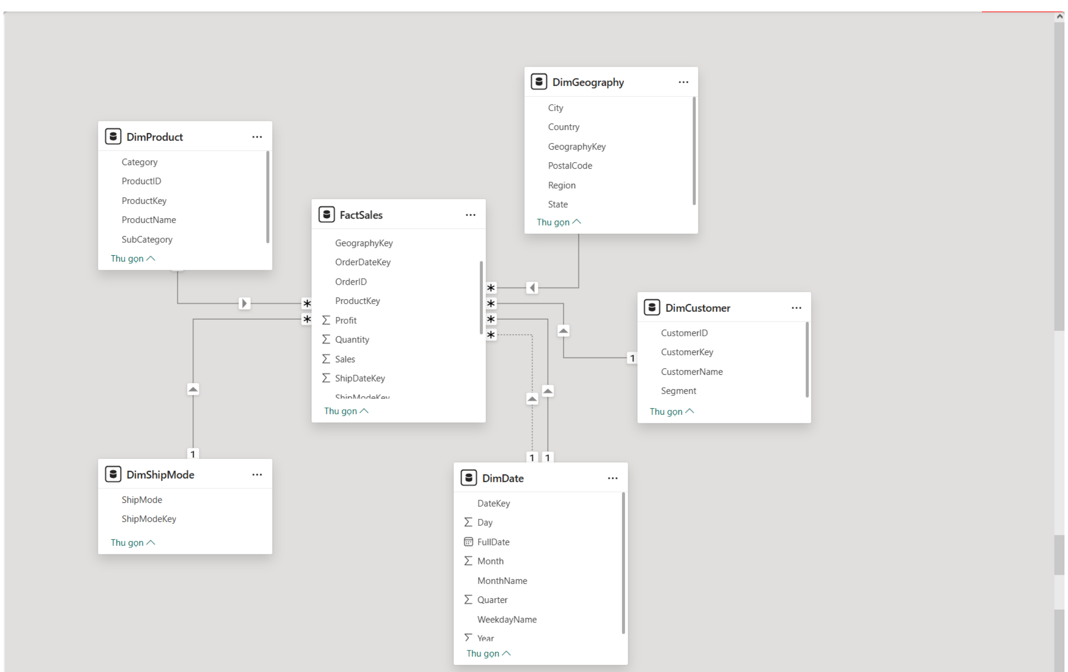


Figure 19: Power BI model view confirming the star schema: `FactSales` at the centre with one-to-many relationships to all five dimension tables.

The Power BI report consists of three pages covering all ten analytical questions.

Page 1 — Overview Dashboard (Figure 20) presents KPI cards (Total Sales, Total Profit, Total

Orders), a bar chart of Sales and Profit by Year, a line chart of Year-over-Year Sales growth, a map of Sales by State, and a top-10 product table.

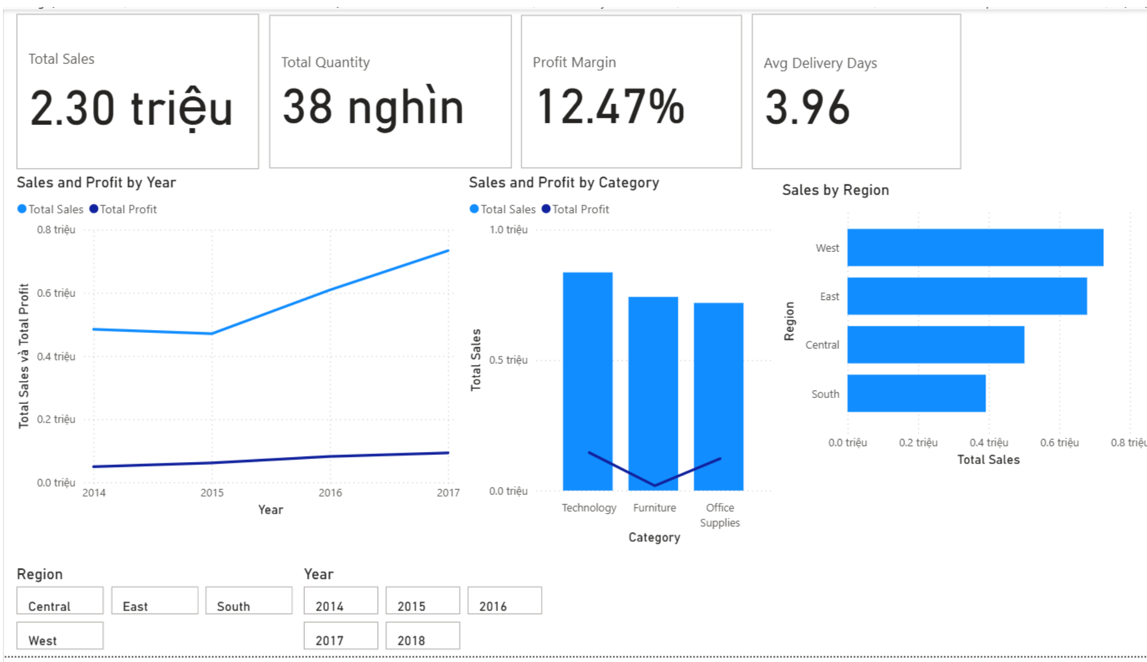


Figure 20: Power BI Overview Dashboard. KPI cards summarise total Sales, Profit, and order count. Bar and line charts track annual trends; a choropleth map highlights Sales by State.

Page 2 — Product and Customer Analysis (Figure 21) shows Sales and Profit by Sub-Category, Profit Margin by Segment, a scatter plot of Discount vs. Profit, and a top-10 customer table.

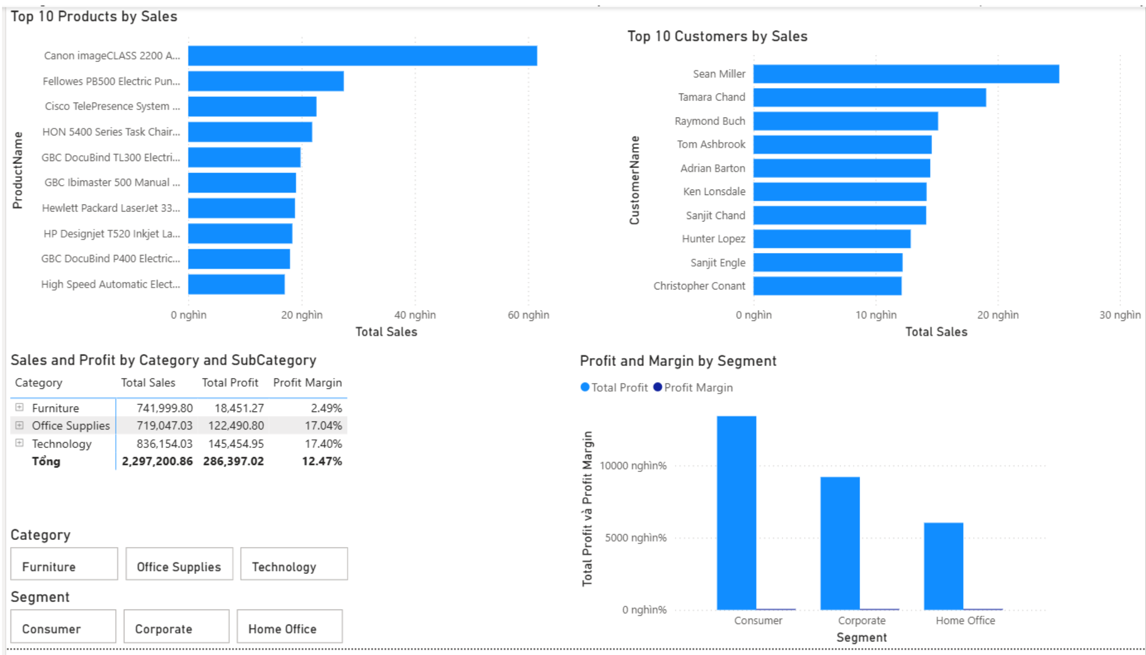


Figure 21: Power BI Product and Customer Analysis page. The scatter plot confirms the negative correlation between Discount and Profit.

Page 3 — Shipping and Geography Analysis (Figure 22) presents Sales and order count by Ship Mode, a bar chart of average delivery days by Region and Ship Mode, and a regional Sales breakdown.

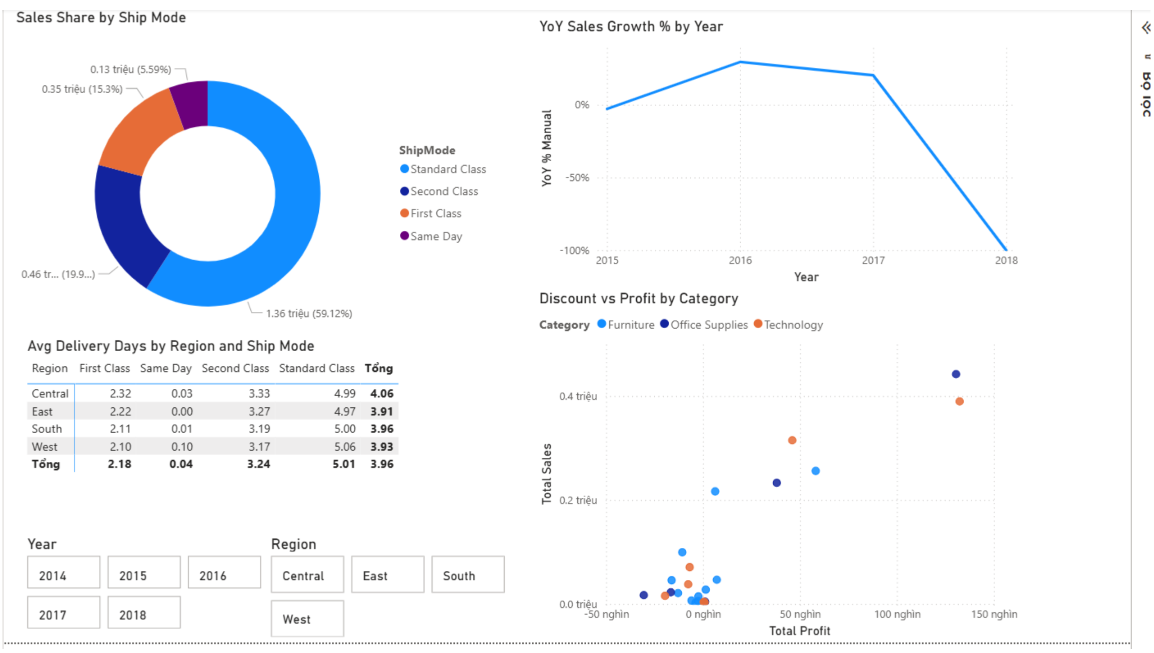


Figure 22: Power BI Shipping and Geography page. Standard Class carries the highest order volume; Same Day shows near-zero average delivery days.

4.3 Excel PivotTable Analysis

Excel connects to the SSAS cube via an Analysis Services OLAP connection, enabling direct drag-and-drop of measures and dimension hierarchies into PivotTables. Four PivotTables are presented.

PivotTable 1 — Sales and Profit by Category and Region (Figure 23) cross-tabulates the three product categories against the four sales regions. Technology leads Sales in all regions; Furniture shows a negative Profit contribution in the East and South regions, reinforcing the discount-driven loss pattern.

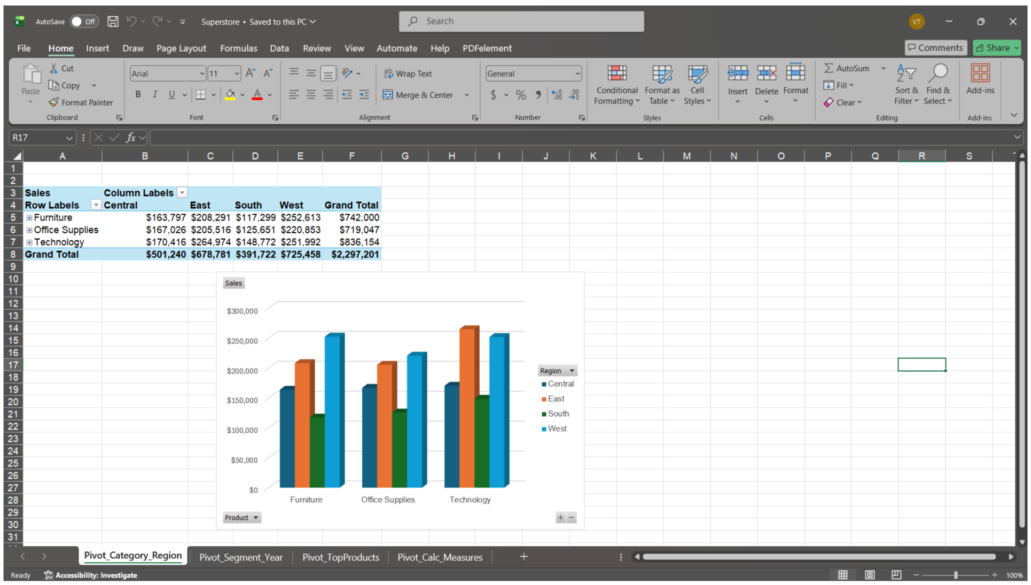


Figure 23: Excel PivotTable 1: Sales and Profit by Category and Region. Furniture records negative Profit in several regions.

PivotTable 2 — Sales by Segment and Year (Figure 24) tracks how each customer segment's Sales evolve annually. The Consumer segment is the largest in absolute Sales throughout; all three segments show year-on-year growth.

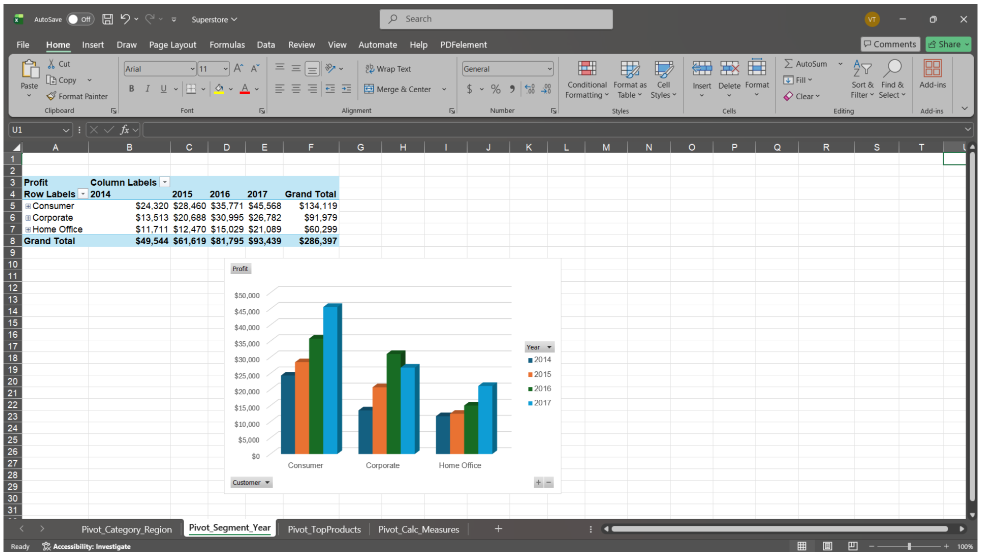


Figure 24: Excel PivotTable 2: Sales by Customer Segment and Year. Consumer consistently leads; all segments grow year-on-year.

PivotTable 3 — Top Products by Sales (Figure 25) lists the highest-revenue products, confirming the MDX Q3 result: Technology items (copiers and phones) dominate.

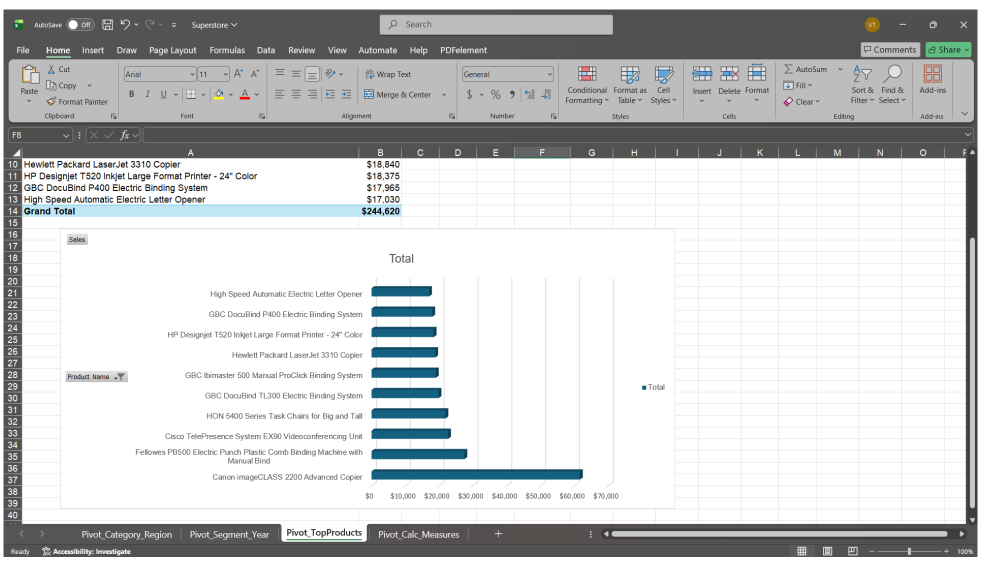


Figure 25: Excel PivotTable 3: Top products by Sales. Copiers and Phones lead, consistent with the MDX Q3 result.

PivotTable 4 — Calculated Measures (Profit Margin and Avg Delivery Days) (Figure 26) retrieves the cube's calculated members directly into Excel, demonstrating that Excel can consume SSAS calculated measures without replicating the formula logic in the spreadsheet itself.

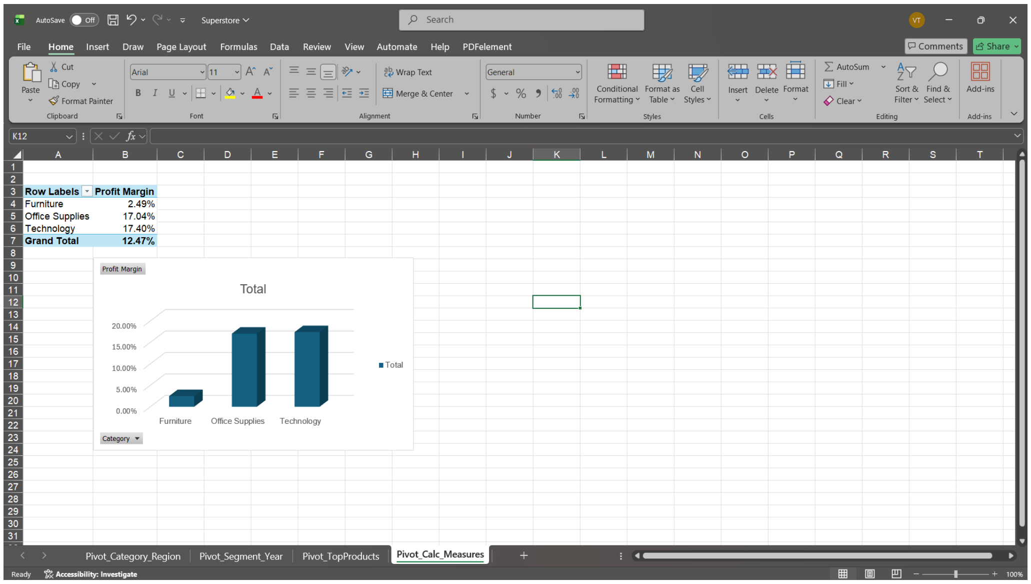


Figure 26: Excel PivotTable 4: Profit Margin and Average Delivery Days from the SSAS calculated members, consumed directly by the Excel OLAP connection.

5 Data Mining

Figure 27 shows the overall data mining workflow. The Python script `DataMining/data_mining.py` applies two algorithms to the Superstore source data: a supervised Decision Tree classifier for loss-risk rule discovery and an unsupervised K-Means model for RFM customer segmentation.

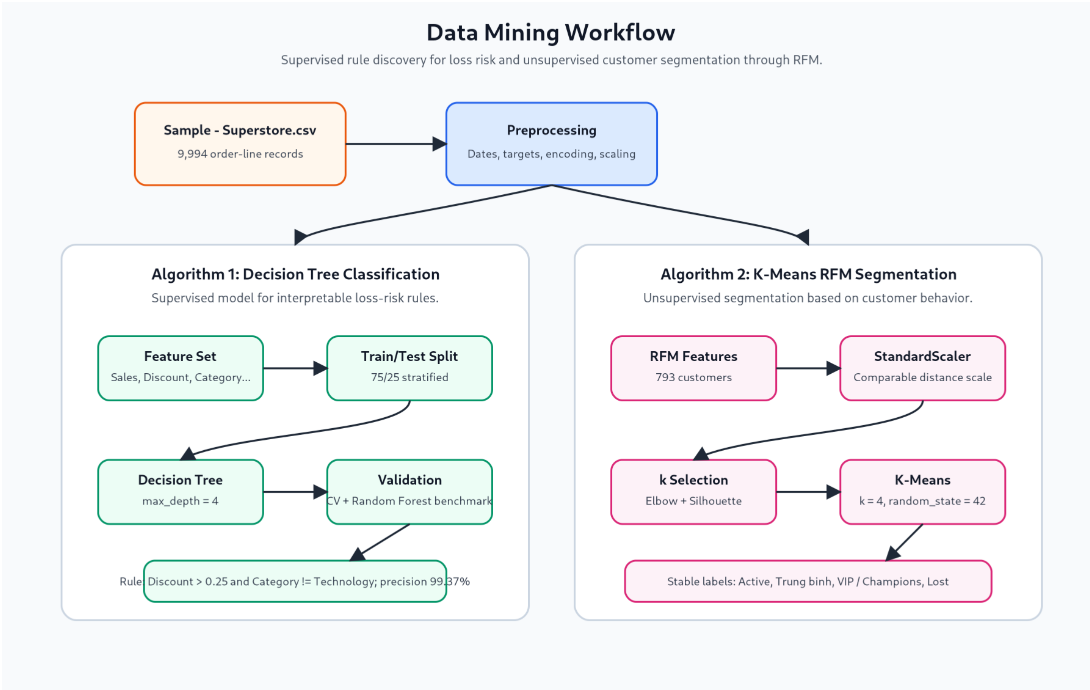


Figure 27: Data mining workflow. The Decision Tree is a supervised classification model; K-Means is an unsupervised segmentation model. Both operate on the Superstore source data.

5.1 Decision Tree — Loss-Risk Rule Discovery

5.1.1 Objective and Feature Set

The binary classification target is defined as:

$$y = \begin{cases} \text{Profitable} = 1 & \text{if } \texttt{Profit} > 0 \\ \text{Loss} = 0 & \text{if } \texttt{Profit} \leq 0 \end{cases}$$

Predictors include three numeric variables (`Sales`, `Quantity`, `Discount`) and five one-hot-encoded categorical variables (`Category`, `Sub-Category`, `Region`, `Segment`, `Ship Mode`).

5.1.2 Class Imbalance

The target is moderately imbalanced (Figure 28): 80.63% of order lines are profitable and only 19.37% are loss-making. This imbalance is important for result interpretation: a naïve predictor that labels every record as “Profitable” would already reach 80.63% accuracy. Any evaluation claim must therefore go beyond accuracy alone and report per-class metrics.

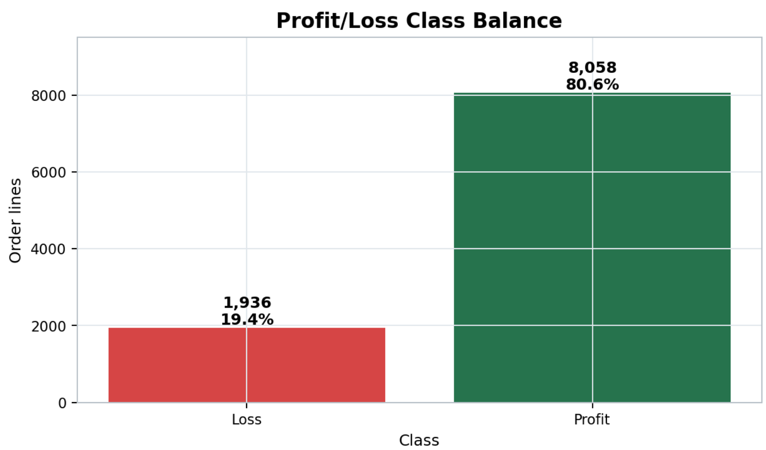


Figure 28: Target class distribution: 80.6% Profitable vs. 19.4% Loss. The imbalance makes macro F1-score a more informative metric than overall accuracy.

5.1.3 Model Performance

The Decision Tree was trained on 75% of the data and evaluated on a held-out 25% test set. Table 5 reports the key metrics. A Random Forest ensemble was also trained as a benchmark; it achieves 94.28% test accuracy, confirming that the Decision Tree's result is not an artefact of model selection.

Table 5: Decision Tree classification performance (held-out test set).

Metric	Value
Total rows	9,994
Class distribution	80.6% Profitable / 19.4% Loss
Baseline accuracy (majority class)	**0.8063**
Decision Tree test accuracy	0.9440
Decision Tree macro F1-score	0.9033
5-fold CV accuracy	0.9444 ± 0.0020
CV loss precision	0.9370
CV loss recall	0.7645
Random Forest benchmark accuracy	0.9428

Source: `DataMining/outputs/model_summary.txt`. *CV = cross-validation.*

The confusion matrix (Figure 29) shows that the model correctly identifies 369 loss-making records and 1,990 profitable records in the test set. It misclassifies 115 loss records as profitable (false negatives) and 25 profitable records as loss-making (false positives).

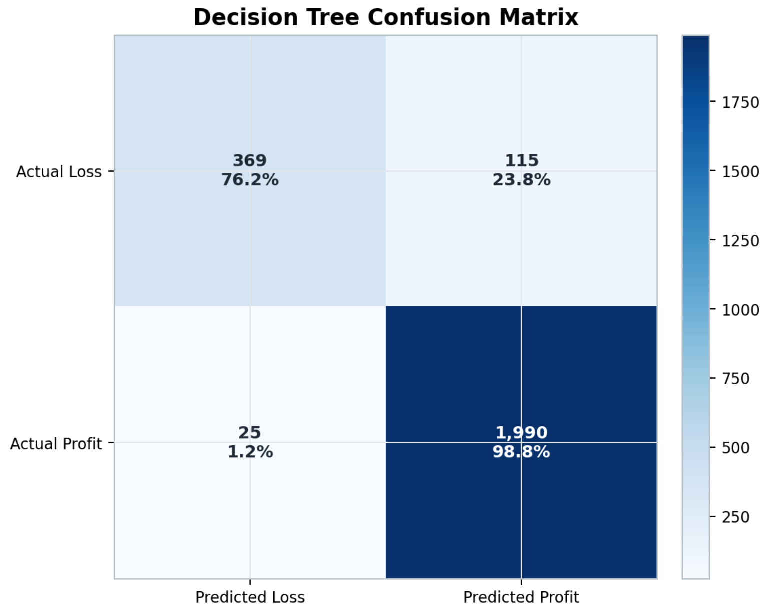


Figure 29: Confusion matrix on the held-out test set. The model correctly identifies 369 out of 484 loss-making records (76.4% recall). False negatives (115 misclassified losses) are the primary limitation.

Business interpretation of the confusion matrix. False negatives (losses predicted as profitable) are operationally more damaging than false positives: a missed loss allows an unprofitable transaction to proceed without review. The 76.4% loss recall reported in cross-validation means the model functions as a high-precision early-warning system rather than a complete loss-prevention gate. It should be used as a decision-support layer that flags suspicious transactions for human review, not as an automated approval engine.

5.1.4 Decision Tree Visualisation and Feature Importance

Figure 30 shows the Decision Tree structure. The root split is on `Discount`, confirming that discounting behaviour is the primary predictor of loss. Figure 31 shows the Random Forest feature importance: `Discount` dominates, with `Sales` and `Quantity` contributing secondary signal; categorical one-hots contribute marginally.

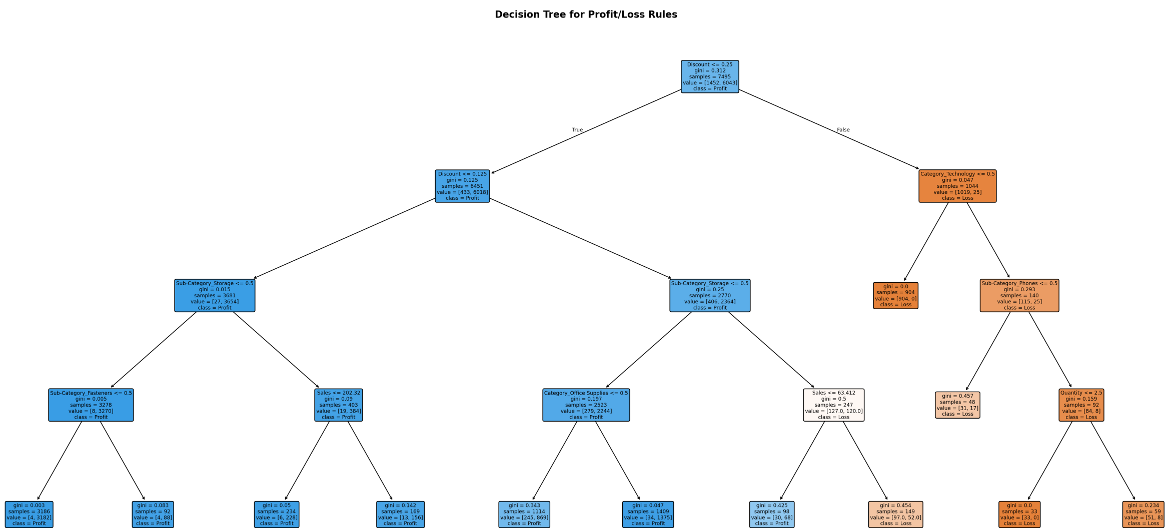


Figure 30: Decision Tree structure (depth limited for readability). The root split occurs on `Discount` at threshold 0.25; subsequent splits refine predictions using `Category` and `Sub-Category`.

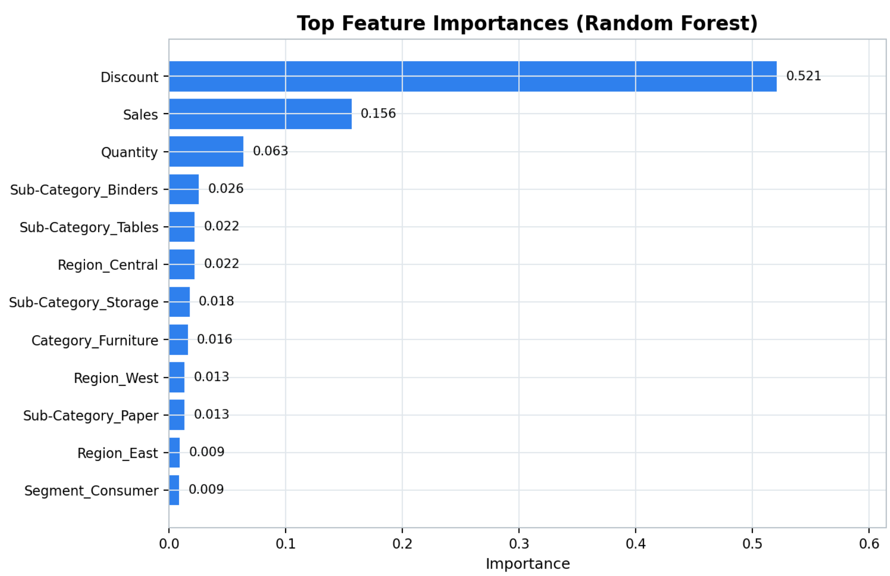


Figure 31: Random Forest feature importance. `Discount` is the dominant predictor; `Sales` and `Quantity` add secondary signal. Categorical one-hots contribute minimal importance.

5.1.5 Validated Business Rule

The script extracts all decision-tree leaf rules and validates each against the held-out test set. Table 6 lists the two loss-predicting rules with the highest test support.

Table 6: Validated loss-predicting decision rules (held-out test set).

Rule	Support	Correct	Incorrect	Precision
`Discount > 0.25 AND Category != Technology`	318	316	2	**99.37%**
`Discount` $\leq$ `0.25 AND Discount > 0.13 AND Sub-Category = Storage AND Sales > $63.41`	45	27	18	60.00%

Source: `DataMining/outputs/loss_rules.csv`.

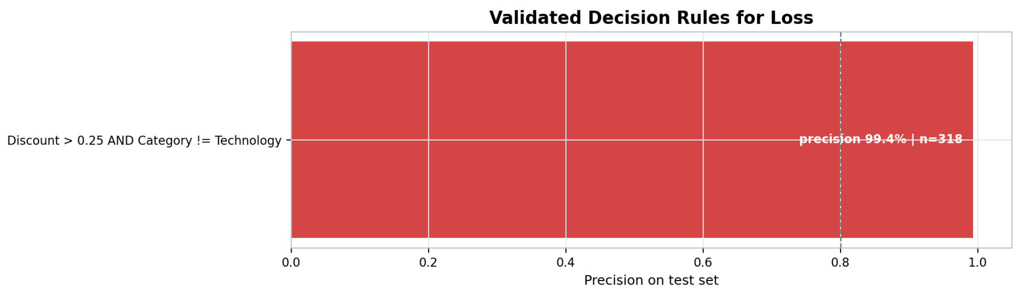


Figure 32: Visualisation of validated loss-predicting rules from the Decision Tree. The primary rule achieves 99.37% precision on 318 test records.

The primary rule: *When the discount exceeds 25% and the product category is not Technology, the order line is very likely to be loss-making (99.37% precision, 318 test records).*

This rule is directly actionable: a sales approval workflow could flag any non-Technology order with a discount above 25% for manager review before the order is finalised. The rule also explains the Furniture Profit deficit observed in MDX Q6—Furniture and Office Supplies carry heavy discounting that consistently destroys margin, while Technology can absorb higher discounts because its unit prices and gross margins are higher.

5.2 K-Means RFM Customer Segmentation

5.2.1 Objective and Preprocessing

The segmentation objective is to group the 793 distinct customers by purchasing behaviour using the RFM framework [3]:

- **Recency (R):** Days since the customer's last purchase (smaller is more recent and more engaged).
- **Frequency (F):** Number of distinct orders placed.
- **Monetary (M):** Total Sales revenue generated by the customer.

Because K-Means uses Euclidean distance, RFM values are standardised with `StandardScaler` before clustering to prevent the Monetary variable's larger absolute scale from dominating distance calculations [4].

5.2.2 Cluster Count Selection

Figure 33 shows the combined Elbow and Silhouette plots. The Elbow curve bends at $k = 4$ and the Silhouette score for $k = 4$ is **0.3554**, indicating moderate cluster separation. A value between 0.25 and 0.50 is typical for real-world customer behaviour data, where purchasing patterns naturally overlap rather than forming perfectly compact clusters [4]. The four-cluster solution was therefore retained.

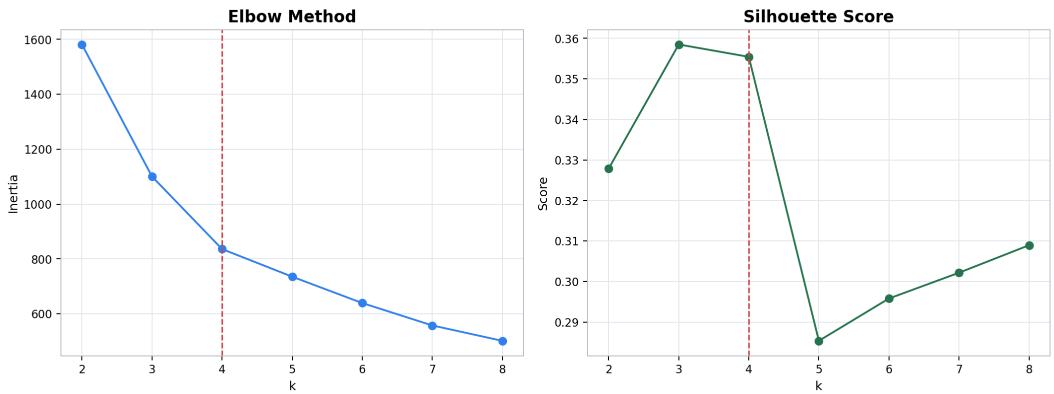


Figure 33: Combined Elbow and Silhouette evidence for cluster-count selection. The Elbow bends at $k = 4$; the Silhouette score at $k = 4$ is 0.3554.

5.2.3 Segmentation Results

Table 7 presents the cluster profiles.

Table 7: K-Means RFM cluster profiles ($k = 4$, Silhouette = 0.3554).

Cluster	Segment Name	Recency	Frequency	Monetary ($)	Count	Proportion
0	Trung thanh / Active	72.7	8.5	3,322.2	298	37.6%
1	Pho thong / Trung binh	101.2	4.7	1,669.7	335	42.2%
2	VIP / Champions	123.7	8.3	9,479.5	64	8.1%
3	Roi bo / Lost	559.5	3.7	1,470.2	96	12.1%

Source: `DataMining/outputs/rfm_profile.csv`. Recency in days (lower = more recent). Monetary based on Sales, not Profit.

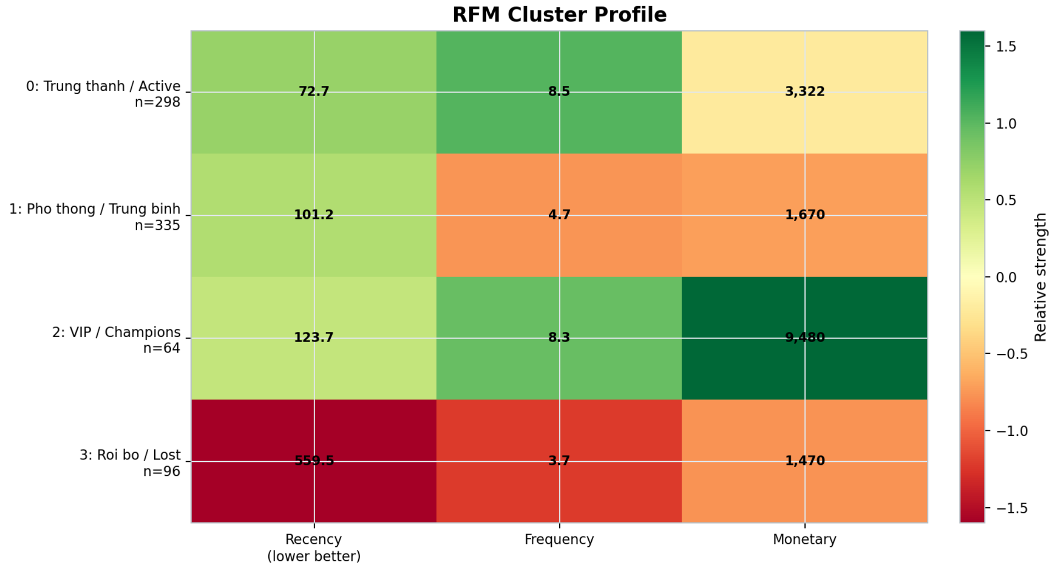


Figure 34: Heatmap of normalised RFM values by cluster. VIP / Champions (Cluster 2) has the highest Monetary; Roi bo / Lost (Cluster 3) has the highest Recency, indicating long inactivity.

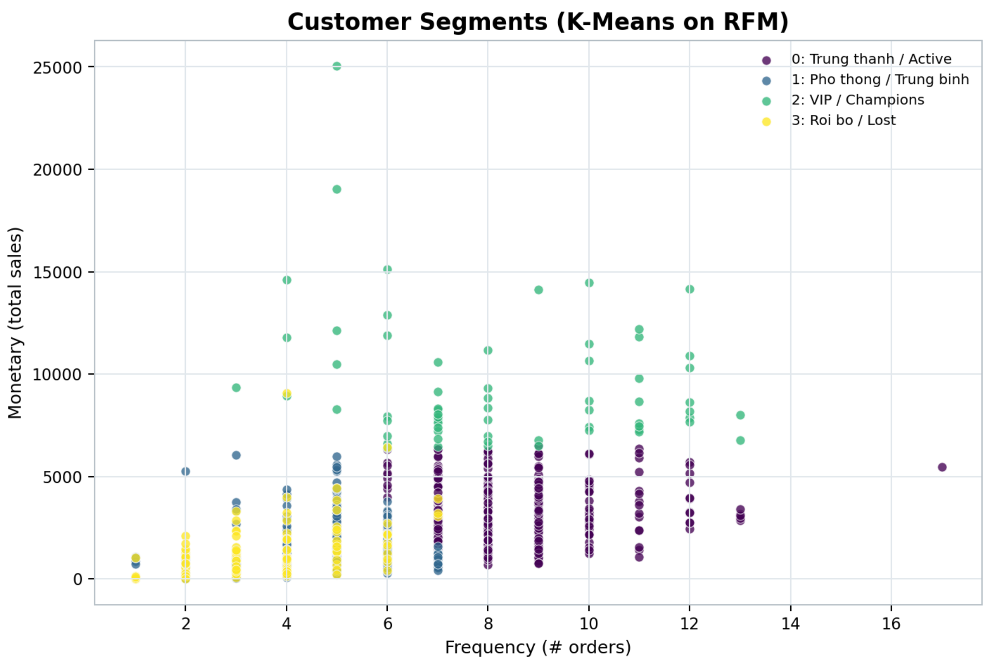


Figure 35: Frequency vs. Monetary scatter plot with segment labels. VIP / Champions occupy the upper-right quadrant (high Frequency, very high Monetary); Roi bo / Lost customers cluster near the lower-left.

5.2.4 Business Interpretation and Recommended Actions

Table 8: Recommended business actions by customer segment.

Segment	Business Interpretation	Recommended Action
VIP / Champions	Highest Monetary value ($9,479.5 average). Only 64 customers but disproportionate revenue contribution. Moderate recency suggests they remain active.	Priority retention: loyalty programme, premium support, early-access product launches, and personalised recommendations. Cost-justify high service investment by revenue share.
Trung thanh / Active	Lowest Recency (72.7 days) and highest Frequency (8.5 orders). Strong engagement with moderate-to-high Monetary value.	Cross-sell and upsell to increase basket size. Target with category- expansion campaigns and referral programmes. Goal: graduate a share of this segment to VIP status.
Pho thong / Trung binh	Largest segment (335 customers, 42.2% of base). Moderate Recency, lower Frequency, lower Monetary. Represents the broad mid-market.	Scalable targeted promotions, bundle offers, and seasonal campaigns. Even modest frequency improvement across 335 customers generates meaningful revenue uplift.
Roi bo / Lost	Very high Recency (559.5 days) indicates long inactivity. Low Frequency and Monetary; likely disengaged or lapsed.	Low-cost win-back campaigns (reactivation email, limited-time discount). Test response rates before committing retention budget. Do not invest in premium retention without evidence of re-engagement.

Methodological note. The Decision Tree and K-Means models are complementary, not redundant. The Decision Tree operates at *order-line* level and produces interpretable loss-risk rules applicable to transaction approval workflows. K-Means operates at *customer* level and produces strategic segmentation applicable to CRM and marketing resource allocation. Together they cover both operational (per-transaction) and strategic (per-customer) decision support.

6 Conclusion

This report has documented an end-to-end Business Intelligence solution for the Superstore dataset, spanning ETL design, dimensional data warehousing, OLAP cube deployment, multi-tool analytical reporting, and Python data mining. The pipeline is fully reproducible: SQL scripts create the database objects, SSIS packages execute the ETL with built-in reconciliation, and the Python script regenerates all data mining outputs deterministically from the source CSV.

Key findings. Four results stand out from the analytical layer:

1. **Discount is the primary loss driver.** The Decision Tree rule `Discount > 25% and Category != Technology` predicts a loss-making order with 99.37% precision on 318 test records. MDX Q6 and Power BI independently confirm that Furniture accumulates negative aggregate Profit, driven by this discount pattern.
2. **VIP customers require disproportionate protection.** The K-Means model identifies 64 VIP / Champions customers ($9,479.5 mean Monetary) whose revenue concentration means that losing even a few to churn has an outsized impact on total revenue.
3. **Technology drives both top-line revenue and margin stability.** MDX Q3 shows Technology items dominating the top-10 product list; Q6 shows Technology maintaining positive Profit even under high discount pressure.
4. **Q4 seasonality is consistent.** Sales peak every Q4 across all available years, suggesting the business should plan inventory and logistics capacity around an annual Q4 demand surge.

Limitations. Three limitations should be noted. First, the Superstore dataset is a benchmark teaching dataset, not live operational data; the time range spans 2014–2017 and does not reflect current market conditions. Second, the ETL package does not implement incremental loading or Slowly Changing Dimension Type 2 logic: a full refresh is required each time the source is updated [1]. Third, the K-Means Silhouette score of 0.3554 indicates moderate rather than strong cluster separation; the segment boundaries should be treated as analytical decision-support labels rather than absolute customer classifications.

Future work. Practical extensions include: scheduling ETL runs via SQL Server Agent for automated incremental refresh; implementing SCD Type 2 in `DimCustomer` and `DimProduct` to track historical attribute changes; deploying a live Power BI Service gateway so dashboards refresh automatically; and applying time-series forecasting (e.g. ARIMA or Facebook Prophet) to the monthly Sales aggregates to produce forward-looking revenue projections.

Appendix A — SQL Scripts

- `sql/01_create_landing_table.sql` — Creates `SuperstoreDW.dbo.stg_Orders`, the internal CSV landing table inside the main `SuperstoreDW` database.
- `sql/02_create_dw.sql` — Creates the five dimension tables and `FactSales` inside `SuperstoreDW`.
- `sql/03_populate_dimdate.sql` — Generates the `DimDate` calendar dimension for the date range required by the dataset.
- `sql/06_verify.sql` — Post-ETL reconciliation: landing/fact row-count comparison,

Sales/Profit total comparison, and orphan foreign-key checks. All checks must pass before proceeding to SSAS deployment.

Appendix B — Python Data Mining

- `DataMining/data_mining.py` — Single script executing the full data mining workflow: data loading, preprocessing, Decision Tree training and evaluation, Random Forest benchmark, rule extraction and validation, K-Means RFM segmentation, Elbow and Silhouette analysis, and all output generation.
- `DataMining/outputs/` — All output files: `PNG` charts, `CSV` tables (`decision_rules.csv`, `loss_rules.csv`, `rfm_profile.csv`, `rfm_customers.csv`), and `model_summary.txt`.
- `tests/test_data_mining_outputs.py` — Automated test suite that verifies output file existence and key numerical assertions after each script run.

Appendix C — SSIS and SSAS Projects

- `SSIS/Integration Services Project1/ETL_Main.dtsx` — Main SSIS package. Contains source reading, dimension loads, and `FactSales` load.
- `SSIS/.../OLEDB_DW.conmgr` — Project-level connection manager pointing to `SuperstoreDW`. Update the server name before deployment on a new machine.
- `SSAS/SuperstoreSSAS/` — SSAS Multidimensional project. The `.dwproj` solution file opens in Visual Studio 2022 with the Microsoft Analysis Services Projects extension.
- `mdx/queries.mdx` — All ten MDX analytical queries with inline comments.

Appendix D — Overleaf Figure Upload Guide

All figures should be uploaded to the `figures/` folder in Overleaf. SVG diagram files must be converted to PNG first (use cloudconvert.com or Inkscape: *File → Export As → PNG at 150 dpi*). Rename them as follows before uploading:

Source file (project folder)	Upload as (in figures/)
Report/diagrams/01-end-to-end-architecture.svg	arch_01_end_to_end.png
Report/diagrams/02-star-schema.svg	arch_02_star_schema.png
Report/diagrams/03-ssis-etl-flow.svg	arch_03_ssis_etl_flow.png
Report/diagrams/04-ssas-cube-architecture.svg	arch_04_ssas_cube.png
Report/diagrams/05-star-vs-snowflake.svg	arch_05_star_vs_snowflake.png
Report/diagrams/06-data-mining-workflow.svg	arch_06_data_mining_workflow.png
Report/screenshots/sql/01_superstoredw_tables.png	01_superstoredw_tables.png
Report/screenshots/sql/02_verify_reconciliation.png	02_verify_reconciliation.png
Report/screenshots/ssis/03_control_flow_success.png	03_control_flow_success.png
Report/screenshots/ssis/04_connection_managers.png	04_connection_managers.png
Report/screenshots/ssis/06_dft_load_factsales.png	06_dft_load_factsales.png
Report/screenshots/ssas/07_data_source_superstoredw.png	07_data_source_superstoredw.png
Report/screenshots/ssas/08_dsv_star_schema.png	08_dsv_star_schema.png
Report/screenshots/ssas/09_cube_browser_sales_profit.png	09_cube_browser_sales_profit.png
Report/screenshots/powerbi/10....png	keep original filename
...all powerbi/ PNGs	keep original filenames
Report/screenshots/mdx/mdx_q*.png	keep original filenames
Report/screenshots/excel/1*.png	keep original filenames
Report/screenshots/datamining/2*.png	keep original filenames

Appendix E — Analytical Question Coverage Matrix

Table 9 maps each original business question to the main evidence used in the report. The purpose of this appendix is to make the submission auditable: every question stated in Section 1 is linked to a concrete analytical artefact and an interpretable business output.

Table 9: Coverage matrix for the ten business questions.

Q	Business question	Primary evidence	Business interpretation
1	Sales and Profit by Year and Quarter	MDX Q1; Power BI overview dashboard; Excel segment-by-year pivot	Revenue grows across the observed period, with Q4 repeatedly acting as the strongest seasonal quarter.
2	Highest Sales and Profit by State	Power BI geography page and map visual	State-level performance identifies where sales concentration and margin contribution should be reviewed together.
3	Top 10 Products by Sales	MDX Q3; Excel Top Products pivot	High-ticket Technology products dominate the revenue ranking and explain the category's top-line strength.
4	Sales and Profit by Product Sub-Category	Power BI product/customer page; Excel category-region pivot	Sub-category analysis separates revenue generators from profit destroyers and supports product-level action.
5	Profit Margin by Customer Segment	MDX Q5; Excel calculated-measure pivot	Corporate leads absolute profit while Home Office shows stronger margin efficiency.
6	Discount effect on Profit by Category	MDX Q6; Power BI discount-vs-profit scatter; Decision Tree rule	Heavy discounting outside Technology is the clearest operational loss-risk signal.

Q	Business question	Primary evidence	Business interpretation
7	Sales volume and order count by Ship Mode	Power BI shipping/geography page	Standard Class carries the largest order volume and should be monitored for service reliability.
8	Year-over-Year Sales growth	MDX Q8; Power BI trend visuals	The cube calculation confirms positive YoY movement across the available time range.
9	Top 10 Customers by Sales	Power BI customer table; RFM segmentation output	Top-customer concentration supports retention investment for high-value accounts.
10	Average delivery time by Region and Ship Mode	MDX Q10; Excel calculated-measure pivot; Power BI shipping page	Same Day performs near zero delivery days; Standard Class is stable at roughly five days.

Appendix F — Submission Evidence Checklist

The checklist below summarises the evidence included for each technical layer. It is intended to support final review before submission and to show that the project is not only a dashboard, but a complete BI pipeline from source data to decision-support interpretation.

Table 10: Technical evidence checklist.

Layer	**Evidence included**	**Validation criterion**
Source data	Dataset description, source-column mapping, encoding note, order-line grain definition	Source contains 9,994 order-line records and the warehouse preserves the same analytical grain.
ETL process	SSIS Control Flow, connection managers, FactSales loading flow, post-load verification	Package uses only `SuperstoreDW`; the internal landing table `stg_Orders` is inside `SuperstoreDW`; no separate staging database is used; row counts and measures reconcile with zero orphan keys.
Data warehouse	Star schema diagram, fact table definition, dimension table summary, star-vs-snowflake comparison	Measures are stored in `FactSales`; descriptive attributes are stored in dimensions; the landing table is internal to `SuperstoreDW`, not a separate database.
SSAS cube	Data source, Data Source View, hierarchies, base measures, calculated members, cube browser screenshot	Cube is processed successfully and supports aggregation by time, product, customer, geography, and ship mode.
MDX analysis	Six representative MDX queries with screenshots and business interpretations	Queries answer core OLAP questions and validate calculated members such as Profit Margin and Average Delivery Days.
Power BI	Source connection, model relationship view, three reporting pages	Report imports only the analytical model and covers overview, product/customer, and shipping/geography analysis.

Layer	Evidence included	Validation criterion
Excel OLAP	PivotTables for category-region, segment-year, top products, and calculated measures	Excel consumes the cube and reuses cube-defined measures without duplicating formulas in the workbook.
Data mining	Decision Tree, Random Forest benchmark, validated rules, K-Means RFM segmentation, output charts	Predictive and segmentation results are evaluated quantitatively and translated into concrete business actions.

Appendix G — Reproducibility and Final Submission Notes

Build Sequence

The recommended build sequence is:

1. Create `SuperstoreDW.dbo.stg_Orders` by running `sql/01_create_landing_table.sql`; then create the dimensional model and calendar dimension with the warehouse SQL scripts.
2. Execute the SSIS package and verify that all Control Flow tasks complete successfully.
3. Run the post-load verification script and confirm that row counts, Sales totals, Profit totals, and orphan-key checks are valid.
4. Deploy and process the SSAS Multidimensional cube.
5. Execute the MDX query file in SSMS and capture query-result evidence.
6. Open the Power BI report and verify the model relationships and report pages.
7. Open the Excel workbook and verify the cube-connected PivotTables.
8. Run the Python data mining script and confirm that all output charts, CSV files, and summary text files are regenerated.

Final Review Notes

- The report should be read as an integrated BI solution, not as isolated screenshots. Each layer depends on the same dimensional warehouse model.
- The most important technical design choice is the star schema centered on `FactSales`. This keeps reporting logic consistent across SSAS, MDX, Power BI, and Excel.
- The ETL uses only `SuperstoreDW`; `stg_Orders` is an internal landing table, not a separate database.
- The most important business result is the discount-loss relationship: discounts above 25% outside Technology should be reviewed before order approval.
- The most important CRM result is the RFM segmentation: VIP / Champions customers should receive retention priority, while Roi bo / Lost customers should be approached with low-cost win-back experiments.
- The data mining models should support managerial judgement rather than replace it. In particular, the Decision Tree has high precision but does not catch every loss-making order line.

References

[1] R. Kimball and M. Ross, *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling*, 3rd ed. John Wiley & Sons, 2013.

[2] Tableau Community / Kaggle Contributors, "Sample superstore dataset," Kaggle dataset, 2018, retail order-line benchmark dataset commonly used for business intelligence and dashboarding practice.

[3] A. M. Hughes, *Strategic Database Marketing*. Probus Publishing, 1994.

[4] A. K. Jain, "Data clustering: 50 years beyond k-means," *Pattern Recognition Letters*, vol. 31, no. 8, pp. 651–666, 2010.